Graph Signal Sampling-Based Active Learning for Hyperspectral Image Classification

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***Abstract*—Active learning (AL) has achieved great success in remotely sensed hyperspectral image (HSI) classification due to its ability to select highly informative training samples. An appropriate query strategy is the core of a successful AL method. In this article, we develop a new graph signal sampling (GS)-based AL strategy that aims to query the most globally optimal and informative pixels for HSI classification. We first demonstrate the significance of smooth graph structures for GS. Then, we propose a multiview fused graph-based GS method (GS-FusG) for AL. Our method combines connection information from multiple perspectives to improve the sampling effectiveness and the performance of graph signal reconstruction (GSR). In addition, we present a new reconstruction method for multidimensional graph signals, which integrates the results of multiview GSR. Last but not least, we develop three variants of GS-FusG based on different multiview compositions and signal reconstruction methods. Our experimental results with real HSIs demonstrate that our proposed GS-FusG can greatly enhance AL effectiveness and classification performance.**

***Index Terms*—Active learning (AL), graph signal sampling (GS), hyperspectral image (HSI) classification, multidimensional graph signal reconstruction (GSR), multiview graph.**

I. Introduction

With the rapid development of deep learning technology [1], [2], [3], [4], current methods for hyperspectral image (HSI) classification need to deal with problems such as the limited number and information of available labeled samples. Normally, the training set is passively determined without the interaction of the classifier, and is then employed to train a classifier that assigns each test pixel with one of multiple classes [5]. The representativeness of the training set significantly influences the performance of the classifier [6], [7], [8]. Intuitively, the larger the training set is, the more representative it tends to be. However, the requirement of a large training set often brings dilemmas in real application scenarios. On the one hand, the labeling procedure is often expensive and time-consuming [9]. On the other hand, large training sets normally introduce redundancy, leading to low computational efficiency. Therefore, it is desirable to have a training set that is as compact and informative as possible.

A. Active Learning (AL) for HSI Classification

AL can actively gather small but useful samples from the viewpoint of classification performance, as opposed to conventional passive learning [5]. The basic idea of AL is to actively query the most informative pixels from the set of unlabeled ones, and then have such pixels labeled by an expert and add them to the training set [10]. Compared with random sampling, classifiers trained on actively labeled samples usually exhibit better predictive performance [11]. In the context of HSI classification, according to the querying strategies adopted, AL methods can be roughly differentiated into uncertainty-based, representativeness-based, the combination of both, query-by-committee, and look-ahead strategies [11], [12], [13].

1) Uncertainty-Based Querying Strategies: Uncertainty-based querying strategies treat the pixels most uncertain to the current classifier as the most informative ones. These pixels are located at the class boundaries predicted by the classifier. So, querying these pixels can enhance the classifier's ability to capture true boundaries of class distributions. Uncertainty can be measured by various statistics, among which Qiu et al. [14] introduced the margin-based measure for support vector machines (SVMs) to query samples closest to the decision boundaries. The prediction entropy (EP) is a measure of uncertainty that can be computed by existing models [15] and is widely used to select the most confusing samples. Breaking ties (BTs), which describes the degree of ambiguity with which a classifier judges the categories of samples, proved to be better than EP [16] and has been preferred by many related works [17], [18], [19]. Mutual information (MI) [20] measures the mutual dependence between the posterior and the class labels. Bayesian AL by disagreement (BALD) [17] seeks the samples with the largest disagreement between the posterior under the parameters and the predicted results. Other methods based on graph convolutional networks (GCNs) [21],

Received 5 May 2025; revised 7 September 2025; accepted 2 October 2025. Date of publication 10 October 2025; date of current version 3 November 2025. This work was supported in part by the National Science Fund for Distinguished Young Scholars of China under Grant T2225019, in part by the National Natural Science Foundation of China under Grant 62571195, in part by Hubei Provincial Natural Science Foundation under Grant 2024AFB105, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515011887, and in part by the Science and Technology Project of Guangzhou Municipal Construction Group Company Ltd. under Grant [2024]-KJ0002. *(Corresponding author: Jun Li.)*

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Digital Object Identifier 10.1109/TGRS.2025.3619985

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[22] learn uncertainties using the associations between labeled and unlabeled samples. However, previous studies [23] have demonstrated that uncertainty-based querying strategies are extremely prone to ignore the aggregated subsets without any labels. On the other hand, they may query samples at biased decision boundaries, disregarding boundaries with fewer labels. All of these factors lead to the degradation of AL performance.

2) Representativeness-Based Querying Strategies: Representativeness-based querying strategies expect to find the labeled set that is most consistent with the distribution of the original data. In cluster-based methods, the samples around the cluster centers are typically taken as the representative subset of the distribution, which are labeled for various classification models such as deep neural networks (DNNs) [24] and GCNs [25]. Similarly, the CoreGCN [21] uses geometric information between the labeled and unlabeled graph representations to pick out the samples that are farthest from the sampling set. The methods based on adversarial learning, either by generating fake features [26] or by reducing the distribution discrepancies between the labeled and the unlabeled set [27], obtain representative samples with low probability of belonging to the labeled set. These samples are less likely to be represented by the labeled set, thus increasing the consistency between labeled and original data distributions. However, the representative sampling set contains insufficient information about the distribution boundary, resulting in blurred decision boundaries.

3) Combination of Uncertainty- and Representativeness-Based Querying Strategies: The strategies combining uncertainty- and representativeness-based methods attempt to ensure that the sampling set can both confuse the classifier and fit the original data distribution. These methods include three frameworks.

1) The unified optimization one considers both uncertainty maximization and representativeness/diversity constraints to jointly solve the optimal sampling set. This framework includes methods such as QUIRE [28] and USDM [29], among others [30].
2) The two-step framework either collects samples at different class boundaries provided by pseudo-labels [19], [20], filters out the diverse subset from uncertain sampling sets by clustering [15], [31], [32], or weights the uncertainty scores and representativeness scores to obtain the new samples [33].
3) The collaborative training framework [34] establishes a collaboration between the original classifier and an additional classifier (trained by representative samples) to improve the performance of uncertainty-based AL.

These methods can be affected by the quality of pseudo-labels or clustering, and thus query unrepresentative or noisy samples. Furthermore, the combination of the two strategies cannot ensure that the selected samples are globally optimal.

4) Query-by-Committee Strategies: Query-by-committee strategies [35], [36], [37] query informative samples that classifiers (in committee) disagree most with. Similarly, multiview AL [38] also trains multiple classifiers based on view-generation methods, such as wavelets [39], multiscale strategies [40], bagging [41], and dynamic view [42]. These approaches, like those based on uncertainty, tend to pay excessive attention to noisy data.

5) Look-Ahead Querying Strategies: Look-ahead strategies leverage the current state to forecast the changes of the model caused by labeling a query sample. The goal of the query functions is to maximize model changes [43], [44], [45] or minimize the estimated risk [46]. However, the optimal solution depends heavily on the quality of the pseudo-labels, which limits the effectiveness of AL under the condition of a small number of labeled samples.

B. Contributions of This Work

The core of AL for HSI classification is the query criterion employed to decide which unlabeled HSI pixels are most informative. The main reason for the limitations in the effectiveness of the above methods is that, in most cases, the querying strategies do not optimally match the classification models. In other words, the acquired samples are not globally optimal. In addition, these methods usually require initial training samples to learn a classifier or distribution in advance, a process that we call *start-up*. Therefore, these methods cannot avoid the random label sampling process, which limits their effectiveness and operability.

To solve these problems, in this work, we propose a new AL method based on a graph signal sampling (GS) querying strategy [47], [48], [49]. In our algorithm, HSI pixels (and their labels) are modeled as a multidimensional graph signal, with labels composing the signal and pixels/superpixels being the vertices of an undirected graph, where the signal lies on. In this undirected graph, the connectivity (edge) among HSI pixels (vertices) can be measured with a variety of weight calculation methods [50]. Our AL strategy is capable of selecting the pixels whose labels are most useful for reconstructing the labels of unlabeled pixels in the sense of strict GS theory. Thus, these determined query pixels are globally optimal with respect to the prediction of unknown labels, naturally avoiding redundancy relative to each other and being of the best diversity. To the best of the authors' knowledge, there is no previous work adopting GS-based AL technology for HSI classification purposes. In addition, we demonstrate theoretically and experimentally that the smoothing of graph structures is beneficial to signal sampling and recovery, which is neglected by traditional GS theory.

The main contributions of our work can be summarized as follows.

1) *HSI Specialized GS-Based AL:* We introduce the GS-based AL algorithm for HSI classification by restricting the sampling range and further propose a new multiview fused graph-based GS method (GS-FusG), which has the potential to preserve the multidimensional connection information and query the globally optimal samples underlying in multiview graphs.
2) *Multidimensional Graph Signal Reconstruction (MD-GSR):* In the theoretical framework of GS, most of the matched graph signal reconstruction (GSR) methods [47], [51], [52] are targeted at 1-D graph signals.

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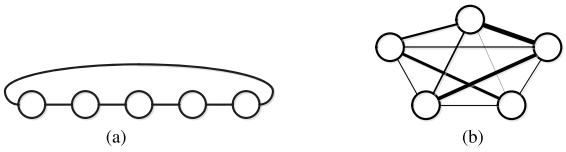


Fig. 1. Illustration of the connectivity of (a) discrete-time signal and (b) graph signal.

We propose a new reconstruction method for multidimensional graph signals (MD-GSR), and the GS-FusG variant based on MD-GSR shows higher AL effectiveness and better HSI classification accuracy. Compared with other classifiers, such as the deep learning-based ones, the mathematical model MD-GSR is of great significance to promote the development of GS theory on multidimensional signals.

3) Our experiments with real HSIs prove that the graph topology has a significant impact on the optimality of GS, and our GS-FusG algorithm outperforms state-of-the-art methods in terms of AL effectiveness and classification performance with limited labels.

The remainder of this article is organized as follows. Section II introduces the background technology of GS theory. Section III describes our new GS-FusG algorithm for HSI classification. Section IV describes the experimental results with real HSIs, which indicate the excellent performance of our method. Section V provides more detailed discussions on the results and characteristics of GS-FusG. Finally, Section VI presents a summary about the advantages of our method, along with some important considerations and suggestions for future AL design.

II. GS Theory

In classical signal processing, a discrete signal refers to a collection of data or samples located on a regular lattice, which is inconsistent with the real world often containing irregular structures and complex relationships [53]. Such irregular structures can be more effectively (and more naturally) represented with graphs. Given an undirected graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ and $\mathcal{E}$ denote the vertex and edge sets, respectively, we can define a graph signal on the graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. Many concepts and tools developed in classical signal processing can be extended to the graph signal processing case [54].

A. Graph Fourier Transform (GFT)

Classical signal processing develops the concept of frequency with the complex exponential $\phi_{k,n} = \exp(j2\pi/Nkn)$. For a finite-length discrete-time signal $\{s[n]\}_{n=0,1,\ldots,N-1}$, its Fourier transform is defined as its projection on N harmonics $\{\phi_{k,\cdot} = \exp(j2\pi/Nkn)\}_{k=1,2,\ldots,N}$ which are a set of complete orthonormal bases of an N-dimensional signal space

$$\bar{s}_k = \langle \mathbf{s}, \phi_{k,\cdot} \rangle_n = \sum_{n=0}^{N-1} s[n] \exp\left(-j\frac{2\pi}{N}kn\right) \tag{1}$$

where $\langle \cdot, \cdot \rangle$ stands for inner production. Considering the discrete-time counterpart ∇_d of the continuous-time Laplace operator $\partial^2/\partial t^2$, its calculation on the nth point of the kth harmonic is

$$\begin{aligned} -\nabla_d \phi_{k,\cdot}[n] &= \left(\phi_{k,n} - \phi_{k,n-1}\right) - \left(\phi_{k,n+1} - \phi_{k,n}\right) \\ &= \left(2 - 2\cos\left(\frac{2\pi}{N}k\right)\right) \exp\left(j\frac{2\pi}{N}kn\right) \\ &= c_k \phi_{k,n} \end{aligned} \tag{2}$$

which can be transformed to the representation of the operation on the whole kth harmonic in a matrix form

$$\mathcal{L}_d \phi_{k,\cdot} = \begin{bmatrix} 2 & -1 & 0 & \cdots & -1 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -1 & 0 & 0 & \cdots & 2 \end{bmatrix} \begin{bmatrix} \phi_{k,0} \\ \phi_{k,1} \\ \phi_{k,2} \\ \vdots \\ \phi_{k,N-1} \end{bmatrix} = c_k \phi_k \tag{3}$$

where the harmonic $\phi_{k,\cdot}$ is the kth eigenvector of the Laplacian matrix $\mathcal{L}_d$ and $c_k = (2 - 2\cos(2\pi/Nk))$ is the associated eigenvalue which reflects the characteristics of frequency. Given the connectivity of discrete-time signal shown in Fig. 1(a), the Laplacian matrix $\mathcal{L}_d$ can be obtained with $\mathcal{L}_d = \mathbf{D}_0 - \mathbf{W}_0$, where $\mathbf{W}_0(i,j) = \begin{cases} 1, & \text{if } i = j \pm 1 \\ 0, & \text{otherwise} \end{cases}$ is the weight matrix corresponding to Fig. 1(a), and $\mathbf{D}_0$ is the degree matrix associated with $\mathbf{W}_0$.

The above relationships can be extended to a graph signal whose samples follow an arbitrary connectivity, as shown in Fig. 1(b). The weight matrix $\mathbf{W}_0$ is then generalized to the weight representation $\mathbf{W}(i,j) = \begin{cases} 0, & \text{if } i = j \\ w_{ij}, & \text{otherwise} \end{cases}$. Similar to (3), the eigenvalue decomposition for a graph signal is

$$\mathcal{L}_n u_k = \lambda_k u_k \tag{4}$$

where the graph Laplacian $\mathcal{L}_n = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$ and the eigenvector u_k denote the kth harmonic for the graph signal, and λ_k is the associated frequency.

Then, the GFT of the graph signal $\mathbf{f} = \{f[n]\}_{n=0,1\cdots N-1}$ can be designed as

$$\hat{\mathbf{f}} = \mathbf{U}^{\mathrm{T}}\mathbf{f} \tag{5}$$

where we have the spectrum $\hat{\mathbf{f}} = [\hat{f}_1, \hat{f}_2, \ldots, \hat{f}_N]^{\mathrm{T}}$ and the harmonic matrix $\mathbf{U} = [u_1, u_2, \ldots, u_N]$.

B. Graph Signal Sampling

For a graph sampling signal $\mathbf{f}(\mathcal{S})$ on the sampling set $\mathcal{S}$, the sufficient and necessary condition for the perfect recovery [55] of an ω-bandlimited signal $\mathbf{f} \in \mathrm{PW}_\omega(\mathcal{G})$ is proved to be

$$\omega < \omega_c(\mathcal{S}) = \inf_{\phi \in L_2(\mathcal{S}^c)} \mathrm{BW}(\phi) \tag{6}$$

where $\omega_c(\mathcal{S})$ is the true cutoff frequency. $L_2(\mathcal{S}^c)$ denotes the null space of the sampling set $\mathcal{S} \subset \mathcal{V}$, i.e., $\forall \phi \in L_2(\mathcal{S}^c)$, $\phi(\mathcal{S}) = 0$. $\mathrm{BW}(\phi)$ is the bandwidth of signal ϕ.

In (6), the spectral moment $\mathrm{SM}_q(\phi) = ((\phi^{\mathrm{T}}\mathcal{L}^q\phi)/(\phi^{\mathrm{T}}\phi))^{1/q} \overset{\phi \in L_2(\mathcal{S}^c)}{=\!=} ((\psi^{\mathrm{T}}(\mathcal{L}^q)_{\mathcal{S}^c}\psi)/(\psi^{\mathrm{T}}\psi))^{1/q}$ can be utilized to approximate $\mathrm{BW}(\phi)$, where $\psi = \phi(\mathcal{S}^c)$, $q \in Z^+$ is the spectral moment order representing the degree of approximation. Then, the cutoff frequency $\omega_c(\mathcal{S})$

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is estimated as $\Omega_q(\mathcal{S}) = \inf_{\phi \in L_2(\mathcal{S}^c)} \mathrm{SM}_q(\phi)$. Furthermore, we have

$$\Omega_q(\mathcal{S}) = \left(\inf_{\phi \in L_2(\mathcal{S}^c)} \frac{\psi^{\mathrm{T}} (\mathcal{L}^q)_{\mathcal{S}^c} \psi}{\psi^{\mathrm{T}} \psi} \right)^{1/q} = \left(\sigma_{1,q}\right)^{1/q} \tag{7}$$

where $(\sigma_{1,q}, \psi_{1,q})$ is the smallest eigen-pair of submatrix $(\mathcal{L}^q)_{\mathcal{S}^c}$ and the smoothest signal ϕ^* can be approximated as $\phi_q^*(\mathcal{S}^c) = \psi_{1,q}$ and $\phi_q^*(\mathcal{S}) = 0$. It has been proven that $\Omega_q(\mathcal{S})$ approximates the real cutoff frequency $\omega_c(\mathcal{S})$ as $q \to \infty$ [55]. But larger values of q lead to more complicated matrix operations $\mathcal{L}^q$. As indicated in (6), the smallest sampling set $\mathcal{S}_{\mathbf{opt}}$ for a given cutoff frequency ω_c can be determined with

$$\begin{aligned} &\min_{\mathcal{S}} \ |\mathcal{S}| \\ &\text{s.t. } \Omega_q(\mathcal{S}) \geq \omega_c. \end{aligned} \tag{8}$$

A greedy heuristic-based selection strategy has been proposed for searching the optimal sampling set $\mathcal{S}_{\mathbf{opt}}$ [49], [51], [52], in which a new sampled vertex v can be calculated by following the criterion:

$$v := \operatorname*{argmax}_i |\phi_q^*(i)| \tag{9}$$

where ϕ_q^* is the qth-order estimation of the smoothest signal in $L_2(\mathcal{S}^c)$ space. Then, the sampling set can be updated until the condition $\Omega_q(\mathcal{S}) \geq \omega_c$ is met

$$\mathcal{S} \leftarrow \mathcal{S} \cup \{v\}. \tag{10}$$

C. 1-D Graph Signal Reconstruction

GSR from sampling portions to unknown parts is essentially an interpolation process [48], which can be implemented with a low-pass filtering strategy. For a 1-D sampling signal $\mathbf{f}(\mathcal{S})$, existing methods [48], [51], [52] recover the original ω-bandlimited signal $\mathbf{f} \in \mathrm{PW}_\omega(\mathcal{G})$ by the following iterative approximation process:

$$\mathbf{f}_0 = \mathcal{P}_\omega \left(\mathcal{D}_{\mathcal{S}}^{-1} \mathbf{f}(\mathcal{S}) \right), \quad \mathbf{f}_{i+1} = \mathcal{P}_\omega \mathcal{P}_{\mathcal{S}} \mathbf{f}_i \tag{11}$$

where $\mathcal{D}^{-1}$ is an upsampling operation, resetting transform $\mathcal{P}_{\mathcal{S}}$ is defined as $\mathcal{P}_{\mathcal{S}} x = x + \mathcal{D}_{\mathcal{S}}^{-1}(f(\mathcal{S}) - \mathcal{D}_{\mathcal{S}} x)$. $\mathcal{P}_\omega = \sum_{j=0}^{r} a_j \mathcal{L}^j$ is the vertex domain representation of the low-pass graph filter whose spectral approximation is the Chebyshev polynomial $\sum_{j=0}^{r} a_j \lambda^j$. $\mathcal{D}_{\mathcal{S}}$ is the downsampling operator.

III. Proposed Method

A HSI can be formulated as a graph signal, which comprises a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ and the associated signal. The pixel set $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{m \times n}]$ of a HSI with b bands and $m \times n$ pixels is treated as the vertex set $\mathcal{V}$. By calculating the similarity values of all pixel pairs on a specific metric, we can obtain the weight matrix $\mathbf{W}$ in which the element w_{ij} in row i and column j represents the connection strength between the ith and the jth vertex of the graph signal. Class labels of HSI pixels are one-hot encoded into a membership matrix $\mathbf{F}$ which can be treated as a multidimensional graph signal. Let $Y = [y_1, y_2, \ldots, y_{m \times n}]$ denote their labels, connected to $\mathcal{C}$ classes. Then, the multidimentional graph signal $\mathbf{F}$ can be generated by

$$F_{ij} = \begin{cases} 1, & i \in [1, m \times n], i \in Z^+, j = y_i \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

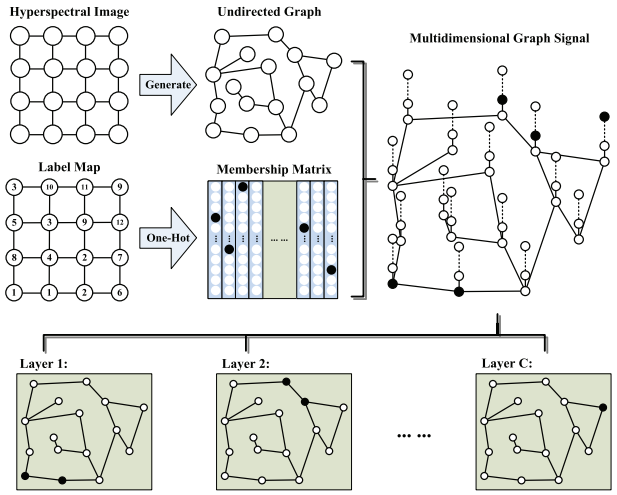


Fig. 2. Multidimentional graph signal constructed by using a HSI.

Fig. 2 provides a graphical example of such a kind of HSI-based graph signals. From the perspective of graph signal processing, HSI classification is essentially a sort of signal recovery procedure from the labeled samples to the unlabeled ones, utilizing the connection weight between the samples of a graph signal. This graph signal recovery or reconstruction process is the inverse of GS, which chooses the most representative or informative samples from a graph signal and, therefore, is truly an AL process.

A. Graph Theory for Sampling and Reconstruction

The GS method can globally locate the samples with the best ability to recover the whole signal. The query method based on graph sampling criteria can naturally obtain the most informative samples with the weakest connections to the training set. However, existing methods only analyze the effect of the sampling method on graph signal recovery. Thus, we first prove the optimality of graph theory for GS and perfect reconstruction.

Without loss of generality, taking $r = 1$ as an example, (11) can be rewritten as

$$\begin{cases} \hat{f}(\mathcal{S}) = f(\mathcal{S}), \\ \hat{f}_{i+1}(\mathcal{S}^c) = a_1 \mathcal{L}_{(\mathcal{S}^c, \mathcal{S})} f(\mathcal{S}) + \left(a_0 \mathbf{I} + a_1 \mathcal{L}_{(\mathcal{S}^c, \mathcal{S}^c)}\right) \hat{f}_i(\mathcal{S}^c). \end{cases}$$

The solution of the above formula is

$$\hat{f}(\mathcal{S}^c) = a_1 \left((1 - a_0) \mathbf{I} - a_1 \mathcal{L}_{(\mathcal{S}^c, \mathcal{S}^c)} \right)^{-1} \mathcal{L}_{(\mathcal{S}^c, \mathcal{S})} f(\mathcal{S}). \tag{13}$$

To simplify the representation, we treat $\mathcal{S}^c$ as the unlabeled sample set and $\mathcal{S}$ as the labeled sample set. Therefore, (13) can be simplified as follows:

$$\hat{f}_u = a_1 \left((1 - a_0) \mathbf{I} - a_1 \mathcal{L}_{uu} \right)^{-1} \mathcal{L}_{ul} f_l. \tag{14}$$

We first analyze the case of a 1-D graph signal, i.e., $f = F_{\cdot j}, j = 1, 2, \ldots, \mathcal{C}$. In $F_{\cdot j}$, signal points belonging to class j have a value of 1, and those not belonging to class j have a value of 0. The optimal graph theory of 1-D graph signal and its proof are as follows.

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Theorem 1: For any sampling set $\mathcal{S}$ that contains at least one signal point belonging to class j, the "smoothest" graph that guarantees perfect recovery of $F_{\cdot j}$ needs to meet the following two requirements.

1) There is no connection between signal points of different classes.
2) There is no "isolated" (i.e., no connection with other signal points in class j) subset of signal points belonging to class j in $\mathcal{S}^c$.

Proof:

1) We divide $\hat{f}_u$ in (14) into two parts

$$\hat{f}_u = \begin{bmatrix} \hat{f}_{u_j} \\ \hat{f}_{u_o} \end{bmatrix}$$

where u_j denotes the subset belonging to class j in the unlabeled set, u_o denotes the subset belonging to other classes in the unlabeled set. Thus, (14) can be rewritten as

$$\begin{bmatrix} (1-a_0)\mathbf{I} - a_1\mathcal{L}_{u_ju_j} & -a_1\mathcal{L}_{u_ju_o} \\ -a_1\mathcal{L}_{u_ou_j} & (1-a_0)\mathbf{I} - a_1\mathcal{L}_{u_ou_o} \end{bmatrix} \begin{bmatrix} \hat{f}_{u_j} \\ \hat{f}_{u_o} \end{bmatrix} = a_1 \begin{bmatrix} \mathcal{L}_{u_jl} \\ left\{\mathcal{L}_{u_ol} \end{bmatrix} f_l.$$

The solution of $\hat{f}_{u_o}$ is

$$\hat{f}_{u_o} = a_1\tilde{\mathcal{L}}\left(u_j, u_o\right)^{-1}\left(\mathcal{L}_{u_ol} + a_1\hat{\mathcal{L}}_{u_ou_j}\mathcal{L}_{u_jl}\right) f_l \tag{15}$$

where $\tilde{\mathcal{L}}(u_j, u_o) = (1-a_0)\mathbf{I} - a_1\mathcal{L}_{u_ou_o} - a_1^2\hat{\mathcal{L}}_{u_ou_j}\mathcal{L}_{u_ju_o}$, and $\hat{\mathcal{L}}_{u_ou_j} = \mathcal{L}_{u_ou_j}[(1-a_0)\mathbf{I} - a_1\mathcal{L}_{u_ju_j}]^{-1}$. From (15), it follows that $\hat{f}_{u_o} = \mathbf{0}$ if and only if $\mathcal{L}_{u_ol} = \mathbf{0}$ and $\mathcal{L}_{u_ou_j} = \mathbf{0}$. In other words, to recover the signal $F_{\cdot j}$ perfectly, we need no connections between signal points of different classes.

2) Let us suppose there is an "isolated" subset u_{s1} belonging to class j in $\mathcal{S}^c$, and u_{s2} is the remaining part of $\mathcal{S}^c$. Then, u_{s1} is not connected to either u_{s2} or $\mathcal{S}$. Equation (14) can be rewritten as

$$\begin{bmatrix} \hat{f}_{u_{s1}} \\ \hat{f}_{u_{s2}} \end{bmatrix} = a_1 \begin{bmatrix} \triangle_{u_{s1}u_{s1}} & \mathbf{0} \\ \mathbf{0} & \triangle_{u_{s2}u_{s2}} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{0} \\ left\{\mathcal{L}_{u_{s2}l} \end{bmatrix} f_l$$

where $\triangle_{ii} = (1-a_0)\mathbf{I} - a_1\mathcal{L}_{ii}$. We derive $\hat{f}_{u_{s1}} = \mathbf{0}$, demonstrating that the graph signal on the isolated subset u_{s1} cannot be recovered. Therefore, to recover the signal $F_{\cdot j}$ perfectly, we do not require an isolated subset of signal points belonging to class j in $\mathcal{S}^c$.

Thus, we prove that graph theory can achieve perfect reconstruction of 1-D graph signals. For multidimensional graph signals, we propose the following theorem.

Theorem 2: The "smoothest" graph perfectly recovering multidimensional graph signal F is one of the following two cases.

1) A multilayer graph, where the jth-layer graph is the one for perfect reconstruction of the jth-dimensional graph signal $F_{\cdot j}$.
2) A single-layer graph that can perfectly recover all dimensions of F.

With the "smoothest" graph, only one sampling point per class can perfectly recover the entire graph signal, so the sampling effectiveness is also maximized. From the above analysis, we can draw a conclusion that the graph has a significant impact on the sampling and recovery of the graph signal. Due to noise, mixtures, and spectral variability in the HSI data, we cannot implement the graphs in Theorem 2. However, each layer of the multilayer graph in the Theorem 2 case 1) corresponds to an independent perspective and is easier to approximate than the single-layer graph in case 2). Therefore, in the following, we develop a GS method based on multiview fused graphs for AL of HSI data, seeking to exploit the characteristics of HSI data to improve the sampling effectiveness and reconstruction performance of graph signals.

B. Multiview Multilayer Graph Construction

For HSI data, the previous graphs normally use a unique similarity metric to calculate the connection weights between pixels, thereby establishing a one-sided local or global structure. These graphs, in a single perspective, cannot build a smooth graph signal from complex HSI data. In this article, we regard AL of HSI data as a signal sampling problem on multilayer graphs, corresponding to case 1) of Theorem 2. We first construct the multilayer graph with different similarity metrics as follows.

1) *D1 (L_2-Norm Distance-Based Radial Basis Function (RBF) [56]):*

$$W_{ij}^{(\mathrm{D1})} = \exp\left(-\frac{\|v_i - v_j\|_2^2}{2\sigma^2}\right) \tag{16}$$

where $v_i, v_j \in \mathcal{V}$ denote the two certain vertices in the vertex set $\mathcal{V}$.

2) *D2 (Feature Mean-Residual L_2-Norm Distance-Based RBF):*

$$W_{ij}^{(\mathrm{D2})} = \exp\left(-\frac{\|(v_i - v_{fi}) - (v_j - v_{fj})\|_2^2}{2\sigma^2}\right) \tag{17}$$

where $v_{fi} = \begin{cases} \sum_{k=1}^{\mathrm{M}} v_i(k)/M, & M > 1 \\ 0, & M = 1 \end{cases}$, and M denotes the feature dimension of the vertex $v_i \in \mathbb{R}^{1\times M}$, $v_i(k)$ denotes the kth element in v_i.

3) *D3 (L_1-Norm Distance-Based RBF):*

$$W_{ij}^{(\mathrm{D3})} = \exp\left(-\frac{\|v_i - v_j\|_1^2}{2\sigma^2}\right). \tag{18}$$

4) *CR1 (Feature Mean-Residual Normalized Correlation):*

$$W_{ij}^{(\mathrm{CR1})} = \left|\frac{(v_i - v_{fi})(v_j - v_{fj})}{\|v_i - v_{fi}\|_2\|v_j - v_{fj}\|_2}\right|. \tag{19}$$

5) *CR2 (Inner Product):*

$$W_{ij}^{(\mathrm{CR2})} = |v_i \cdot v_j|. \tag{20}$$

6) *CR3 (Cosine Distance):*

$$W_{ij}^{(\mathrm{CR3})} = \left|\frac{v_i \cdot v_j}{\|v_i\|_2\|v_j\|_2}\right|. \tag{21}$$

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7) *CR4 (Sample Mean-Residual Normalized Correlation):*

$$W_{ij}^{(\mathrm{CR4})} = \left| \frac{(v_i - v_s)(v_j - v_s)}{\|v_i - v_s\|_2 \|v_j - v_s\|_2} \right| \tag{22}$$

where $v_s = \begin{cases} \sum_{k=1}^{\mathrm{N}} v_k/N, & N > 1 \\ 0, & N = 1 \end{cases}$ and N is the number of vertices.

8) *CV1 (Feature Mean-Residual Covariance):*

$$W_{ij}^{(\mathrm{CV1})} = \left| \frac{(v_i - v_{fi})(v_j - v_{fj})}{N-1} \right|. \tag{23}$$

9) *CV2 (Sample Mean-Residual Covariance):*

$$W_{ij}^{(\mathrm{CV2})} = \left| \frac{(v_i - v_s)(v_j - v_s)}{N-1} \right|. \tag{24}$$

In the multilayer graph, each similarity metric can be taken as a view. Thus, the multilayer graph contains connective information from different perspectives. In terms of connection mode, traditional GS methods usually adopt the k-nearest neighbor graph (KNG), which is defined by

$$w_{ij}^{k(\cdot)} = \begin{cases} w_{ij}^{(\cdot)}, & \text{if } v_i \in \mathcal{N}^k(v_j) \text{ or } v_j \in \mathcal{N}^k(v_i) \\ 0, & \text{otherwise} \end{cases} \tag{25}$$

where $\mathcal{N}^k(v_i)$ is the k-nearest neighbor set of v_i, calculated by vertex features. $w_{ij}^{(\cdot)}$ is the weight under a specific similarity metric, where $(\cdot)$ can be assigned as (D1), ..., (CV2), and so on.

In HSI data, KNG essentially assumes that pixels with similar spectral signatures likely belong to the same class. However, due to noise, mixtures, and spectral variability, KNG brings a lot of mal-connections. To suppress the influence of mal-connections, we jointly exploit the spatial information (which is ignored by existing GS methods) and the spectral information contained in HSI pixels to construct the spatial nearest neighbor graph (SNG) as follows:

$$w_{ij}^{S(\cdot)} = \begin{cases} w_{ij}^{(\cdot)}, & \text{if } v_i \in \mathcal{N}^S(v_j) \\ 0, & \text{otherwise} \end{cases} \tag{26}$$

where $\mathcal{N}^S(v_j)$ is the spatial nearest neighbor set of v_j. The spatial information in SNG is manifested in two aspects: 1) the sparse edges in (26) explicitly represent the spatial adjacency relationships among pixels and 2) SNG-based reconstruction implicitly imposes a spatial smoothness constraint.

Then, we combine different similarity metrics and connection modes (KNG and SNG) to form the multilayer graph with the edge vector denoted as

$$\mathbf{e}_{ij} = \left[w_{ij}^{k(\mathrm{D1})}, \ldots, w_{ij}^{k(\mathrm{CV2})}, w_{ij}^{S(\mathrm{D1})}, \ldots, w_{ij}^{S(\mathrm{CV2})} \right]. \tag{27}$$

Each element in the edge vector $\mathbf{e}_{ij}$ represents an independent view. Among them, different similarity metrics increase the connection information from multiple metric spaces, while SNG accounts for spatial information and KNG models long-range dependences. Therefore, our newly developed multilayer graph contains rich connection information from different views and has the ability to approximate the smoothest graph in Theorem 2.

C. Shared Graph Encoding

Since vector-valued edges cannot be used directly for graph signal processing, we encode the connection information in the edge vectors into a single-layer graph. On the one hand, we note that the single-layer graph of case 2) in Theorem 2 can be obtained by fusing all layers of the multilayer graph in case 1). On the other hand, due to the large number of layers, weighted fusion will introduce too many parameters. Therefore, we develop a dimensionality reduction method to adaptively fuse all layers of the multilayer graph.

First of all, considering the difference in the degree matrix of the graph in different layers, we modify the edge vector of (27) into a normalized form, as follows:

$$\mathbf{e}_{ij} = \left[\bar{w}_{ij}^{k(\mathrm{D1})}, \ldots, \bar{w}_{ij}^{k(\mathrm{CV2})}, \bar{w}_{ij}^{S(\mathrm{D1})}, \ldots, \bar{w}_{ij}^{S(\mathrm{CV2})} \right] \tag{28}$$

where each element of $\mathbf{e}_{ij}$ is normalized by $d_{ii}^{k(\cdot)} = \sum_{j=1}^{\mathrm{N}} w_{ij}^{k(\cdot)}$ or $d_{ii}^{S(\cdot)} = \sum_{j=1}^{\mathrm{N}} w_{ij}^{S(\cdot)}$, mathematically denoted as

$$\bar{w}_{ij}^{k(\cdot)} = \frac{w_{ij}^{k(\cdot)}}{\sqrt{d_{ii}^{k(\cdot)}}\sqrt{d_{jj}^{k(\cdot)}}}, \bar{w}_{ij}^{S(\cdot)} = \frac{w_{ij}^{S(\cdot)}}{\sqrt{d_{ii}^{S(\cdot)}}\sqrt{d_{jj}^{S(\cdot)}}}.$$

Let us assume that $\mathbf{e}_{ij}$ consists of N_v views. Then, we construct an edge matrix $\tilde{\mathbf{E}} \in \mathbb{R}^{N^2 \times N_v}$ containing all the edges of our newly developed multilayer graph

$$\tilde{\mathbf{E}} = \left[\mathbf{e}_{11}^{\mathrm{T}}, \ldots, \mathbf{e}_{N1}^{\mathrm{T}}, \ldots, \mathbf{e}_{1N}^{\mathrm{T}}, \ldots, \mathbf{e}_{NN}^{\mathrm{T}} \right]^{\mathrm{T}}.$$

Building upon existing dimensionality reduction and information extraction techniques [57], [58], such as the widely adopted principal component analysis (PCA), we propose a novel dimensionality reduction method for the edge matrix $\tilde{\mathbf{E}}$, formulated as the following optimization problem:

$$\max_{\substack{\mathbf{P}^{\mathrm{T}}\mathbf{P}=\mathbf{I} \\ \mathbf{P} \geq \mathbf{0}}} \mathbf{P}^{\mathrm{T}} \tilde{\mathbf{E}}^{\mathrm{T}} \mathbf{H} \tilde{\mathbf{E}} \mathbf{P} \tag{29}$$

where $\mathbf{P} \in \mathbb{R}^{N_v \times 1}$ and $\mathbf{H} = \mathbf{I} - 1/N^2 \mathbf{1} \in \mathbb{R}^{N^2 \times N^2}$. Equation (29) maximizes the variance of shared graph weights while constraining the elements of the encoding matrix $\mathbf{P}$ to be nonnegative to prevent the generation of negative connections. To simplify this problem, we let $\tilde{\mathbf{E}}$ contain enough $\mathbf{0}$ vector-valued edges to ensure that the constraint $\mathbf{P} \geq \mathbf{0}$ is satisfied. Then, the problem (29) can be solved by the Lagrange multiplier method, which gives

$$\tilde{\mathbf{E}}^{\mathrm{T}} \mathbf{H} \tilde{\mathbf{E}} \mathbf{P} = \mathbf{P} \lambda_0 \tag{30}$$

where λ_0 is the largest eigenvalue and $\mathbf{P}$ is the corresponding eigenvector. We use $\mathbf{P}$ as the learned encoding matrix to get a 1-D representation of the edge matrix as $\tilde{\mathbf{E}}^0 = \tilde{\mathbf{E}}\mathbf{P}$. Thus, the final weight matrix of our newly developed fused graph (FusG) is

$$\mathbf{W}^{(\mathrm{FusG})} = \begin{bmatrix} \tilde{\mathbf{E}}_1^0 & \cdots & \tilde{\mathbf{E}}_{(N-1)\times N+1}^0 \\ \vdots & \ddots & \vdots \\ \tilde{\mathbf{E}}_N^0 & \cdots & \tilde{\mathbf{E}}_{N^2}^0 \end{bmatrix} \tag{31}$$

where $\tilde{\mathbf{E}}_i^0$ denotes the ith element in $\tilde{\mathbf{E}}^0$. This fused graph $\mathbf{W}^{(\mathrm{FusG})}$ is shared by the multidimensional graph signal for sampling and reconstruction.

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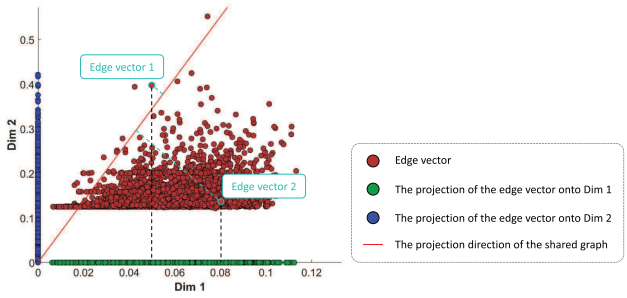


Fig. 3. Example diagram of a set of 2-D edge vectors.

An advantage of fusing graphs through (29) is that the variance of edge features is maximized, thus maintaining the diversity of edges and preserving the multidimensional connection information in the edge vectors. Another advantage is that the fused edge weights can be learned automatically from all edge vectors without the need to set additional fusion parameters. As shown in the 2-D case in Fig. 3, when these vectors are projected onto the direction of the shared graph, their weights can be adjusted based on information from both dimensions.

Our newly proposed FusG not only considers multiple similarity metrics to optimize graph connections but also utilizes spatial information while ensuring long-term dependences through the fusion of KNG and SNG. Thus, FusG has the potential to approximate the smoothest graph in Theorem 2, so as to improve the sampling effectiveness of graph signals and reduce the reconstruction error.

D. GS-FusG-Based AL

Our newly proposed GS-FusG and the subsequent classification procedure are described in Fig. 4. First, for an HSI, the FusG is constructed as a shared graph through (31). Next, the high-order graph Laplacian matrix of FusG is calculated as

$$\mathcal{L}_n^q = \left(\mathbf{I} - \hat{\mathbf{D}}^{-1/2}\mathbf{W}^{(\mathrm{FusG})}\hat{\mathbf{D}}^{-1/2}\right)^q \tag{32}$$

where $\hat{\mathbf{D}}_{ii} = \sum_j \mathbf{W}_{ij}^{(\mathrm{FusG})}$. With $\mathcal{L}_n^q$ and the sampling set $\mathcal{S}^{(t)}$ in the tth iteration, GS-FusG calculates the eigenvector ψ_1' corresponding to the smallest eigenvalue of $(\mathcal{L}_n^q)_{\mathcal{S}^{c(t)}}$ and gets the FusG-based smoothest graph signal ϕ', with $\phi'(\mathcal{S}^{c(t)}) = \psi_1'$, $\phi'(\mathcal{S}^{(t)}) = 0$, and $\mathcal{S}^{c(t)}$ being the complementary set of $\mathcal{S}^{(t)}$. Then, GS-FusG performs the following sampling criterion:

$$\begin{cases} v := \underset{i}{argmax}\,|\phi'| \\ \mathcal{S}^{(t+1)} := \mathcal{S}^{(t)} \cup \{v\},\ \mathcal{S}^{c(t+1)} := \mathcal{S}^{c(t)} \setminus \{v\}. \end{cases} \tag{33}$$

In the subsequent $(t+1)$th iteration, the algorithm repeats: 1) eigen-decomposition of $(\mathcal{L}_n^q)_{\mathcal{S}^{c(t+1)}}$ and 2) sampling with (33). Through the above iterative update of $\mathcal{S}$, the proposed GS-FusG has the potential to collect the globally optimal points underlying in the multiview graph structures, due to the fusion characteristics of FusG illustrated in Fig. 3.

At the expert labeling stage, different from the generalized GS framework that treats the entire $\mathcal{S}$ as the labeled set, our GS-FusG further utilizes the candidate set to limit the range of active labeling. In our framework, the candidate set $\mathbf{D}_C$ is defined as a pool of partial unlabeled graph vertices, with preallocated spatial positions. Therefore, we obtain the labeled set $\mathbf{D}_L$ by labeling the intersection of the sampling set and the candidate set

$$\mathbf{D}_L \leftarrow \mathcal{S} \cap \mathbf{D}_C. \tag{34}$$

Algorithm 1 GS-FusG

Input:
An HSI with the pixel set $\mathbf{X}$, the initial labeled set $\mathbf{D}_L^{(0)}$ and the candidate set $\mathbf{D}_C^{(0)}$; The number of queries N_q, spectral moment order $q \in Z^+$.

1: **Initialization:** Construct FusG using $\mathbf{X}$ and Eq. (31); calculate the high-order Laplacian matrix $\mathcal{L}_n^q$.
2: **Repeat:**
3: Compute the eigen-decomposition of $(\mathcal{L}_n^q)_{\mathcal{S}^{c(t)}}$ for $\mathcal{S}^{(t)}$ and get the smoothest signal ϕ';
4: Sample the vertex v and update the sampling set using Eq. (33);
5: **if** $v \in \mathbf{D}_C$ **then**
6: Label the vertex v and refresh $\mathbf{D}_L$ and $\mathbf{D}_C$ using

$$\mathbf{D}_L^{(t+1)} := \mathbf{D}_L^{(t)} \cup \{v\}; \tag{35}$$

$$\mathbf{D}_C^{(t+1)} := \mathbf{D}_C^{(t)} \setminus \{v\}; \tag{36}$$

7: **end if**
8: **Until** $|\mathbf{D}_L^{(t+1)}| = |\mathbf{D}_L^{(0)}| + N_q$

Output: The final labeled set $\mathbf{D}_L^* := \mathbf{D}_L^{(N_q)}$.

In the HSI scenario, (34) can prevent a large number of uninteresting background pixels in the HSI from being queried, thus improving AL effectiveness. The detailed process of GS-FusG is shown in Algorithm 1. Finally, with $\mathbf{D}_L^*$, GS-FusG calculates the estimated cutoff frequency for reconstruction as

$$\Omega_q\left(\mathbf{D}_L^*\right) = \left(\sigma_1''\right)^{1/q} \tag{37}$$

where σ_1'' is the smallest eigenvalue of $(\mathcal{L}_n^q)_{\bar{\mathcal{S}}^c}$ and $\bar{\mathcal{S}} = \mathbf{D}_L^*$.

Compared with the existing HSI AL methods, our proposed GS-FusG has two advantages.

1) Sampling Effectiveness: GS-FusG essentially queries the globally optimal samples in multiview graphs, specifically tailored for multidimensional graph signals in the real HSI scene, overcoming the limitation of conventional GS techniques that overlook graph topology's influence on optimality.

2) Model-Free AL Process: Most of the AL methods are model-dependent, meaning they require a repetitive cycle of "query–labeling–model updating." While GS-FusG is independent of the classification model, it can start querying samples with no initial labels. As shown in Fig. 5, due to the necessity of model updating, the traditional AL methods can only follow a fixed path during the process of collecting labels on-site, thus incurring higher costs. On the contrary, our proposed GS-FusG can calculate all the points to be labeled at once, thus enabling path planning in field exploration. In addition, based on GS-FusG, we can consider the cost of expert labeling and the

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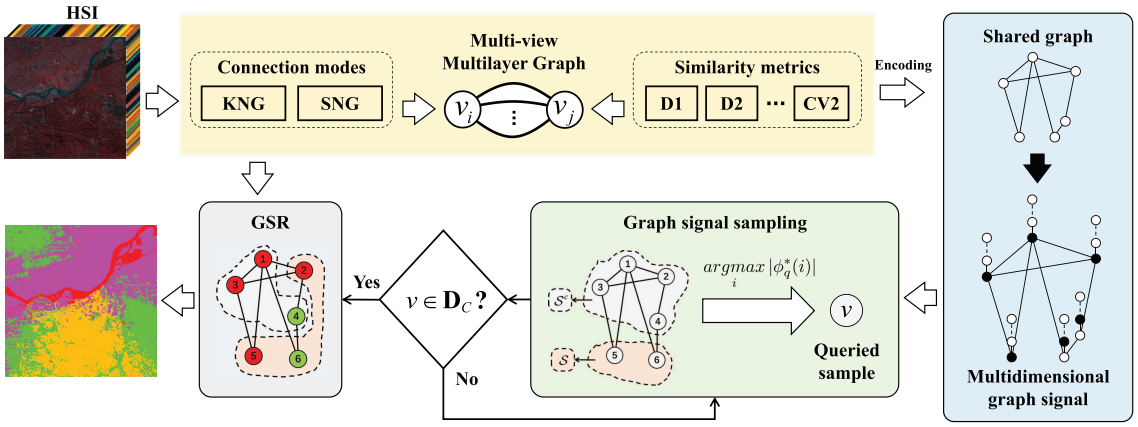


Fig. 4. Flowchart of our newly proposed AL method (GS-FusG). First, multiview multilayer graphs are generated by performing diverse connection modes and similarity metrics on HSI data. Second, we encode the multidimensional connection information of multilayer graphs into a shared graph (FusG). By combining FusG and GS technique, we proposed the GS-FusG method—a new AL framework for HSI, and developed a novel MD-GSR algorithm for HSI classification.

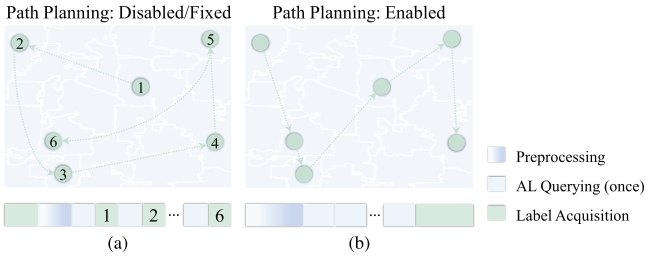


Fig. 5. Advantage of the proposed GS-FusG in the AL process of HSI. (a) Traditional AL. (b) GS-FusG-based AL.

characteristics of the geographical region to decide whether to label these queries one by one.

E. Multidimensional GSR

Since the graph signal $\mathbf{F}$ is encoded by the HSI labels through (12), the GSR process essentially performs classification on HSI, which perfectly matches the AL strategy of our proposed GS-FusG. Here, we present two classification methods based on GSR for HSIs.

1) *1D-GSR:* The original 1D-GSR operates on $\mathbf{f} \in \mathbb{R}^{N\times 1}$ using (11). For the multidimensional graph signal $\mathbf{F} \in \mathbb{R}^{N\times C}$ in the HSI (C is the number of classes), we perform the 1D-GSR operation for each dimension of $\mathbf{F}$ using the shared graph and the cutoff frequency $\Omega_q(\mathbf{D}_L^*)$ and replace $\mathcal{S}$ in (11) with $\mathbf{D}_L^*$.
2) *MD-GSR:* We further consider that the shared graph generated by (31) may have information loss, and cannot achieve effective reconstruction on each dimension of $\mathbf{F}$ at the same time. Therefore, we propose a new multidimensional GSR method to reduce information loss in multiple views, as shown in Fig. 6(c). Suppose that the GSR result under the ith view is $\hat{\mathbf{F}}^{(i)}$; then, the final recovered multidimensional graph signal is

$$\hat{\mathbf{F}} = \hat{\mathbf{F}}^{(1)} \circ \hat{\mathbf{F}}^{(2)} \circ \cdots \circ \hat{\mathbf{F}}^{(K)} \tag{38}$$

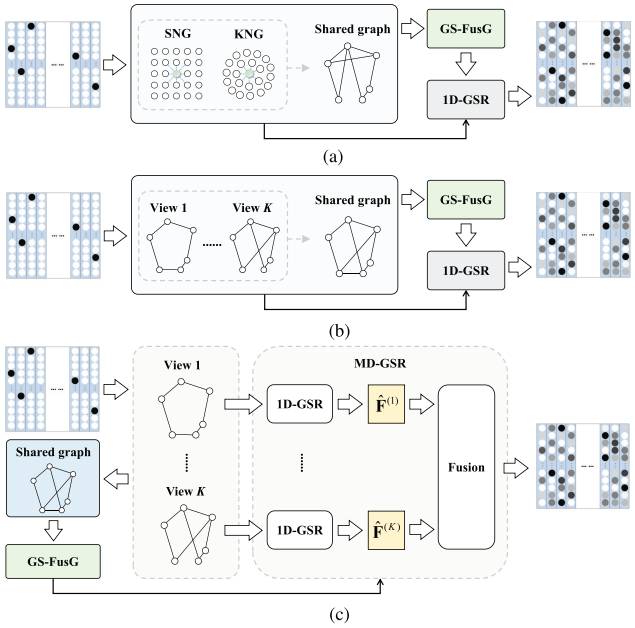


Fig. 6. Three variants of the proposed GS-FusG, in which (a) GS-FusG^1 queries vertices based on FusG from dual views, (b) GS-FusG^2 collects globally optimal vertices underlying in multiview graphs, and (c) GS-FusG^3 fuses the classification results of multiple views.

where $\circ$ denotes the elementwise product and K is the total number of views.

Finally, we give three variants of the GS-FusG method as follows.

1) *GS-FusG^1:* It constructs FusG from dual-view (SNG and KNG on two specific metrics) and performs 1D-GSR, as shown in Fig. 6(a).
2) *GS-FusG^2:* It produces FusG through all the views and performs 1D-GSR, as shown in Fig. 6(b).
3) *GS-FusG^3:* It generates FusG through all the views and adopts MD-GSR, as shown in Fig. 6(c).

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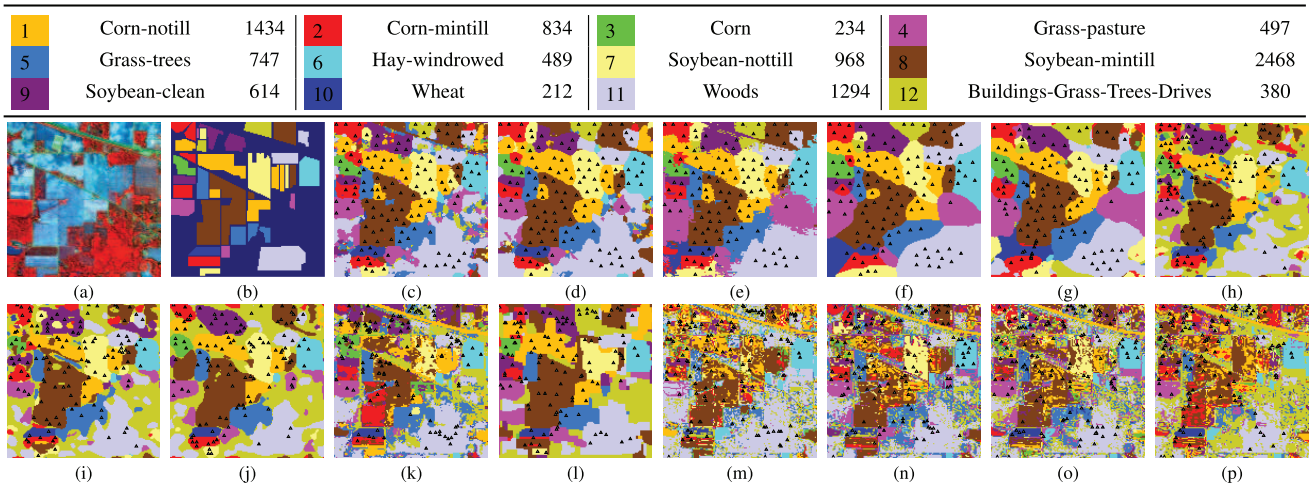


Fig. 7. Classification maps obtained after applying different AL algorithms with 124 labeled samples (illustrated in white triangles) to the Indian Pines dataset. (a) False color composition. (b) Ground truth. (c) GS-FusG[1] (proposed). (d) GS-FusG[2] (proposed). (e) GS-FusG[3] (proposed). (f) GS-SNG. (g) GS-KNG. (h) MI-MLR. (i) BT-MLR. (j) CASSL-MLR. (k) PLCJ. (l) CNN-AL-MRF. (m) MC-MGR. (n) VOPT-MGR. (o) SOPT-MGR. (p) UNC-MGR.

IV. Experiments

This section discusses the prominent capability of our newly proposed GS-FusG for HSI classification by experimental comparison with some commonly used AL algorithms (MI-MLR and BT-MLR) [20], semi-supervised AL algorithms (CASSL-MLR) [34], recent deep learning-based AL algorithms (PLCJ [25] and CNN-AL-MRF [15]), graph-based AL algorithms (PLCJ, MC-MGR [45], SOPT-MGR [59], VOPT-MGR [46], UNC-MGR [45], GS-SNG, and GS-KNG), and some relevant graph-based classification methods (NSCKL [60] and GraphGST [61]). In these methods, PLCJ is based on GCN and adopts the representativeness-based querying strategy, while MC-MGR uses the look-ahead querying strategy. Furthermore, considering that SNG and KNG are two commonly used graph structures in most existing graph-based methods, we implemented GS-SNG and GS-KNG as traditional GS-based methods. In the implementation details of the above algorithm, 3-D Gabor filtering and Gaussian filtering are two alternative preprocessing methods for datasets of different sizes.

In terms of parameter settings, the scale parameter σ and the number of neighbors in the graph are set as the empirical values. Specifically, the number of neighbors adopts fixed values [ten neighbors in $\mathcal{N}^k(v_j)$ and four neighbors in $\mathcal{N}^S(v_j)$]. The spectral moment order q in $\mathcal{L}^q$ is set as $q = 2$ for reduced computational complexity. In the GSR classifier, we set the number of iterations of (11) to 100 (take any large value to ensure the recovery effect). Other parameters are tuned with cross validation.

In our experiments, three benchmark datasets (Indian Pines, WHU-Hi-HanChuan, and WHU-Hi-HongHu) and a newly labeled dataset (Wen-County) were adopted. As the non-GS methods require initial training samples for a *start-up* process, we randomly selected two labeled pixels per class for all the considered HSI datasets, i.e., $2 \times \mathcal{C}$ labeled pixels in total ($\mathcal{C}$ is the number of classes) as the initial training set for these competitors. For the GS-based methods, the active querying and labeling process starts directly with 0 labeled samples. Therefore, when the non-GS methods actively query N_q samples, the GS-based methods can actively label $N_q+2\times\mathcal{C}$ samples.

A. Experiments With AVIRIS Indian Pines Dataset

The AVIRIS Indian Pines dataset consists of 145×145 pixels and 220 bands, including 16 categories of ground objects other than the background, as shown in Fig. 7(a) and (b). We remove 40 bands with numbers 1:5, 101:111, 148:166, and 216:220, which have with low signal-to-noise ratio. On account of the few number of pixels in categories of *Alfalfa*, *Grass-pasture-mowed*, *Oats*, and *Stone-Steel-Towers*, only the remaining 12 categories were used in our experiments and are listed in the table at the top of Fig. 7, including the class names and their corresponding number of pixels.

Using the Indian Pines data, we carried out experiments to evaluate the performance of GS-FusG from the following three perspectives: sensitivity to similarity metrics, AL effectiveness, and classification effect.

1) Performance on Multiple Similarity Metrics: To observe the effect of the graph structure on the GS process, we first compare the performance of the conventional GS method (GS-SNG and GS-KNG) with our proposed GS-FusG on multiple similarity metrics corresponding to (16)–(24). In these metrics, CR2, CV1, and CV2 do not contain normalization, leading to k-nearest neighbors that tend to contain samples with larger features. Thus, these three similarity metrics will not be applied in the KNG-based method.

Tables I–III list the overall accuracies (OAs) obtained when the number of queried/labeled pixels (N_q) reached 54, 84, 124, and 224, respectively. As can be seen from these tables, the choice of a given similarity metric has a significant impact on the GS process; therefore, analyzing the performance on different metrics is crucial. In this experiment, our proposed GS-FusG[1] demonstrated excellent information fusion capability, flexibility, and robustness with multiple metrics, as summarized below.

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TABLE I

OAs (%) Obtained for the Indian Pines Dataset With GS-SNG Using Different Similarity Metrics

Metrics of GS-SNG	N_{cl}	OA (%)			
		$N_q = 54$	$N_q = 84$	$N_q = 124$	$N_q = 224$
CR1	41	**86.68**	**90.52**	92.36	95.77
CR2	209	78.28	84.63	90.08	93.87
CR3	52	82.61	88.02	89.88	95.87
CR4	27	83.22	87.33	92.65	**97.25**
CV1	53	83.11	87.42	90.98	95.41
CV2	81	78.18	88.88	**94.61**	95.98
D1	24	80.50	88.36	91.84	95.42
D2	39	81.31	84.98	89.36	95.66
D3	**20**	81.52	87.11	91.70	94.07
avg.	61	81.71	87.47	91.50	95.48

TABLE II

OAs (%) Obtained for the Indian Pines Dataset Using GS-KNG With Different Similarity Metrics

Metrics of GS-KNG	N_{cl}	OA (%)			
		$N_q = 54$	$N_q = 84$	$N_q = 124$	$N_q = 224$
CR1	35	81.93	88.50	92.69	95.61
CR3	51	86.11	88.42	**93.85**	95.42
CR4	26	85.93	**89.57**	90.98	95.36
D1	33	85.69	87.73	91.92	95.01
D2	26	**87.48**	89.31	91.58	95.31
D3	**23**	85.35	89.45	92.37	**96.67**
avg.	32	85.42	88.83	92.23	95.56

TABLE III

OAs (%) Obtained for the Indian Pines Dataset Using GS-FusG[1] With Different Similarity Metrics

Metrics of GS-FusG[1]	N_{cl}	OA (%)			
		$N_q = 54$	$N_q = 84$	$N_q = 124$	$N_q = 224$
CR1-CR1	34	86.64	91.03	92.30	96.47
CR3-CR3	**19**	88.42	91.85	93.55	96.74
CR4-CR4	**19**	85.20	88.13	90.96	96.34
D1-D1	29	87.63	90.44	92.98	96.82
D2-D2	27	**89.73**	**92.60**	93.67	96.88
D3-D3	24	88.58	90.83	93.35	96.48
CR4-D2	27	88.11	91.36	**95.55**	**97.39**
CR1-CR3	33	87.87	91.30	93.24	96.72
CR1-D3	22	88.21	90.62	92.94	97.13
avg.	26	87.82	90.91	93.14	96.77

1) *Fusion Capability:* GS-FusG[1] is superior to GS-KNG and GS-SNG with each corresponding similarity metric. This reflects that GS-FusG[1] can fuse complementary information in GS-KNG and GS-SNG.
2) *Flexibility:* Both GS-KNG and GS-SNG can only carry out AL with a single metric, while GS-FusG[1] can integrate the information of different similarity metrics (see Table III, where CR4-D2 represents the fusion of SNG on CR4 and KNG on D2) to obtain higher classification accuracies. The classification performance of GS-FusG[1] on CR4-D2 is better than that of GS-KNG and GS-SNG on any metric.

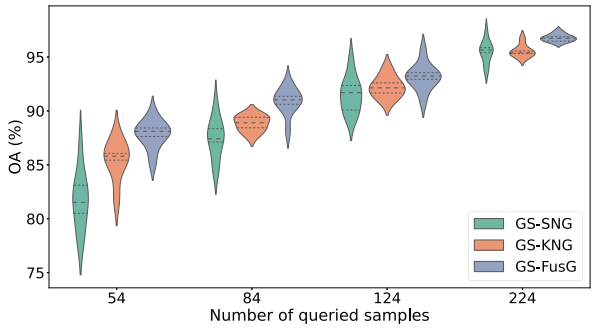


Fig. 8. Sensitivity analysis of GS-based method using similarity metrics.

3) *Robustness:* After comparing the "avg." (average value) in these tables, we can conclude that GS-FusG[1] obtains the best average results, reflecting its best overall performance. Moreover, we employed a violin plot (see Fig. 8) to visualize the distribution of the results on multiple similarity metrics. As can be seen, the results of GS-FusG[1] are densely distributed in regions with high OA values and exhibit low variance. These observations prove that the proposed GS-FusG[1] is more robust than GS-KNG and GS-SNG, owning the lowest sensitivity to similarity metrics.

In addition to OAs, we propose to evaluate the performance of GS-based methods with the number of samples (N_{cl}) in the minimum complete set. We define the concept of a minimum complete set as follows.

Definition 1: The GS-based methods start querying new samples with 0 initial labels. When the labeled set contains all the categories to be predicted, this set is named a complete set. The complete set with the smallest number of samples is defined as the minimum complete set.

A smaller N_{cl} indicates that GS-based methods can obtain the minimum complete set only with fewer querying iterations, that is, GS can make the classifier obtain all classes of ground object information more quickly. In Table III, GS-FusG[1] requires only 19 labeling steps at most, and its N_{cl} values are concentrated in a small range (26 ± 5), with the lowest mean value compared with the competitors. The N_{cl} of GS-SNG is 61 ± 55 (up to 209), and that of GS-KNG is 32 ± 9. GS-FusG[1] achieves the smallest standard deviation of N_{cl}, further highlighting its low sensitivity to similarity metrics.

2) Performance in Terms of AL Effectiveness: We evaluated AL effectiveness from two aspects.

1) The first aspect to be analyzed is the number of queries required by the algorithms to obtain the minimum complete set (i.e., N_{cl}). Tables I–III show that GS-FusG[1] can obtain a minimum complete set with the smallest N_{cl} in terms of both average value and optimal similarity metric. Therefore, GS-FusG[1] exhibits the highest sampling effectiveness in extracting the minimum complete set and providing complete prior label information to the classifier.
2) Second, we assessed the gain in classification accuracy during AL. In this part (and subsequent experiments), the

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TABLE IV

EXPERIMENTAL DETAILS (OA [%], AA [%], AND KAPPA) ON THE AVIRIS DATASET WITH TOTAL 124 LABELED SAMPLES

classNo.	Non-graph-based method		Graph-based method						
	CASSL-MLR	CNN-AL-MRF	PLCJ	VOPT-MGR	NSCKL	GraphGST	GS-FusG^1	GS-FusG^2	GS-FusG^3
1	81.40 ± 6.86	86.47 ± 4.97	39.81 ± 18.70	46.61 ± 11.17	90.46 ± 5.44	94.70	93.36	96.04	**97.24**
2	70.91 ± 11.31	65.17 ± 22.17	44.84 ± 13.46	46.65 ± 8.86	80.87 ± 3.93	87.01	**93.67**	83.74	91.75
3	82.11 ± 17.66	69.94 ± 12.60	69.37 ± 22.44	43.27 ± 11.80	92.70 ± 9.05	59.31	**97.40**	93.51	84.42
4	83.58 ± 10.62	85.19 ± 10.94	78.83 ± 10.12	86.03 ± 5.54	91.98 ± 7.70	94.09	**95.51**	91.85	92.67
5	96.51 ± 3.19	97.30 ± 1.76	91.79 ± 3.63	66.55 ± 16.29	98.51 ± 1.08	99.19	99.19	99.59	**99.73**
6	99.38 ± 0.62	98.64 ± 1.22	99.34 ± 0.27	98.91 ± 0.31	99.58 ± 0.00	**100.00**	**100.00**	**100.00**	**100.00**
7	81.92 ± 14.09	78.72 ± 12.16	38.30 ± 26.15	65.42 ± 6.45	82.58 ± 7.14	95.38	**98.64**	95.59	96.01
8	89.26 ± 4.71	89.66 ± 8.11	66.75 ± 11.98	64.65 ± 7.43	78.62 ± 4.65	88.69	**96.31**	95.70	95.45
9	74.94 ± 7.67	83.20 ± 10.72	64.05 ± 13.50	41.74 ± 7.25	88.08 ± 5.08	91.43	**93.23**	89.62	87.97
10	91.90 ± 13.22	96.18 ± 3.49	98.00 ± 3.15	60.52 ± 20.74	99.50 ± 0.00	99.04	99.52	**100.00**	99.52
11	84.44 ± 7.33	97.95 ± 1.29	89.51 ± 9.81	87.13 ± 5.41	89.79 ± 9.45	99.38	99.53	99.69	**100.00**
12	97.34 ± 3.01	91.00 ± 4.84	56.26 ± 6.33	37.01 ± 13.20	**97.94** ± 2.79	81.12	69.23	83.78	94.95
OA	85.42 ± 1.80	87.35 ± 1.53	65.46 ± 5.65	63.40 ± 3.12	87.40 ± 1.77	92.40	95.55	94.79	**95.79**
AA	86.14 ± 1.78	86.62 ± 2.77	69.74 ± 3.81	62.04 ± 1.99	90.88 ± 1.65	90.78	94.63	94.09	**94.98**
Kappa×100	83.35 ± 2.02	85.44 ± 1.86	60.42 ± 6.32	58.34 ± 3.17	85.66 ± 1.99	91.30	94.90	94.03	**95.18**

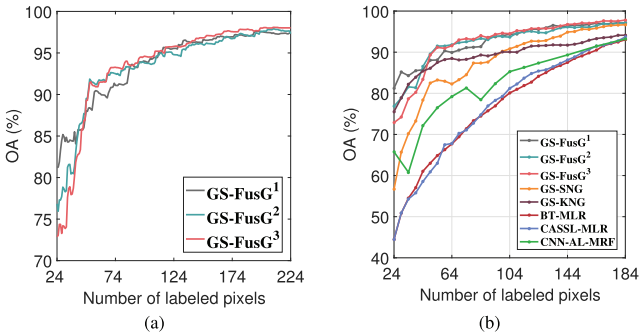


Fig. 9. OA curves obtained for the Indian Pines dataset by (a) three variants of our proposed GS-based method and (b) all AL methods.

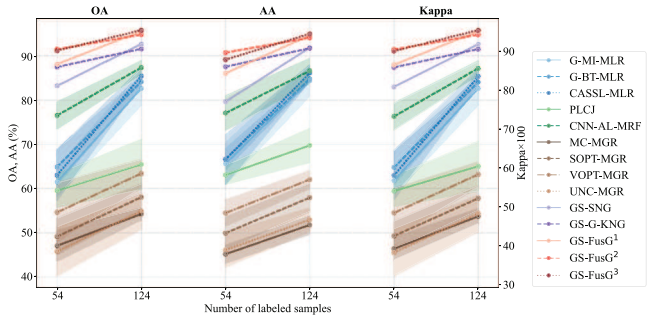


Fig. 10. Comparison of OA (%), AA (%), and Kappa obtained for the AVIRIS Indian Pines by different AL methods.

results of GS-KNG, GS-SNG, and GS-FusG^1 on optimal similarity metrics were compared. As shown in Fig. 9(a), of the three variants of our proposed GS-based method, GS-FusG^1 is the best in the initial iteration phase, while GS-FusG^3 exhibits higher accuracies as the number of labeled samples increases. In Fig. 9(b), our proposed methods consistently outperform other competitors. To achieve OA > 90%, the traditional GS-based methods require 99 labeled samples, whereas our newly proposed HSI-specific GS-based methods only need 59 labeled samples. In comparison, other AL methods require more than 144 labeled samples.

3) Performance in Terms of Classification Accuracy: With a fixed number of labels, we evaluate the classification performance quantitatively by using three indicators: OA, average accuracy (AA), and kappa coefficient (Kappa), and qualitatively by displaying the obtained classification maps. The advantages of our GS-FusG algorithm in classification performance are reflected in the following three aspects.

1) *Stability:* In Fig. 10 and Table IV, all GS-based methods (including our newly proposed GS-FusG method) can get deterministic OAs, AAs, and Kappas, while other methods need random initial labeled samples to train models, so their results are all with standard deviation. Therefore, the performance of our method on real data is stable. (Note: in Table IV, 124 labeled samples indicate that 100 samples were actively queried by non-GS methods.)
2) *Precision:* Fig. 10 shows that our GS-FusG methods achieved the highest accuracies regardless of the number of labeled samples. With a small number of labels, GS-FusG^2 and GS-FusG^3 (using nearly all similarity metrics) have made significant improvements over GS-FusG^1 (using dual similarity metrics). Even with a large number of labels, GS-FusG^3 continues to outperform GS-FusG^1. In Table IV, the proposed GS-FusG variants enhance the classification accuracies across multiple categories with high consistency, holding great superiority over other graph-based methods in recent years. Among them, GraphGST also achieved satisfactory performance (OA >90%) because it uses the same labeled samples as GS-FusG^3. These results demonstrate the validity of our proposed multiview GS strategy.
3) *Segmentation Quality:* In Fig. 7, the classification maps obtained by our methods captured the real boundaries of land covers and also maintained local smoothness, which can best resemble the ground truth. Among them, GS-FusG^1 reconstructed more detailed and accurate boundaries than GS-SNG and GS-KNG through the dual-view GS process; GS-FusG^2 considers the GS

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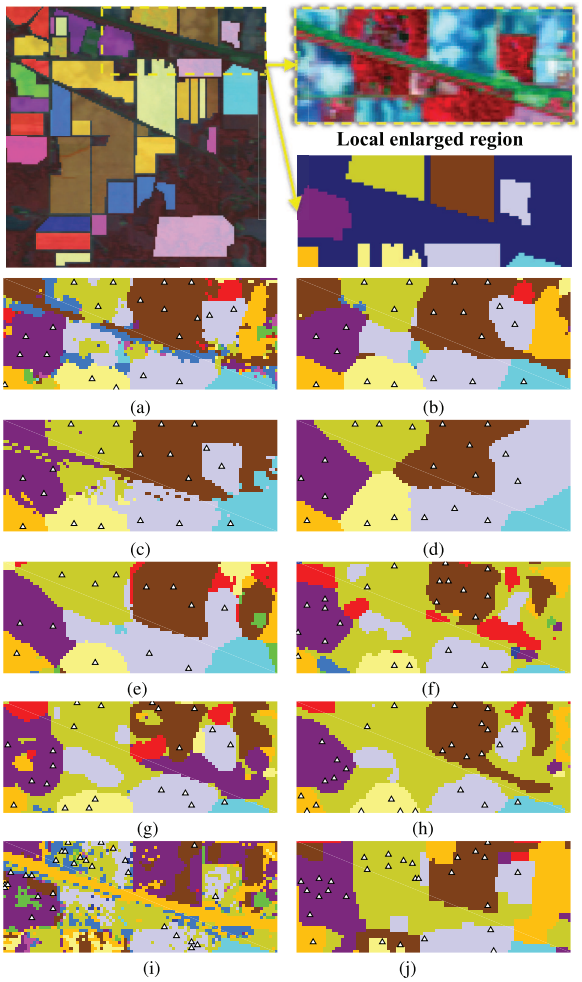


Fig. 11. Segmentation advantage and label distribution (illustrated in white triangles) of the proposed method in local areas. (a) GS-FusG[1] (proposed). (b) GS-FusG[2] (proposed). (c) GS-FusG[3] (proposed). (d) GS-SNG. (e) GS-KNG. (f) MI-MLR. (g) BT-MLR. (h) CASSL-MLR. (i) PLCJ. (j) CNN-AL-MRF.

process in more views, resulting in a smoother map; GS-FusG[3] optimizes the reconstruction of GS through the posterior fusion of multiple views, and obtained a classification map with the highest fidelity in terms of boundaries and smoothness. Furthermore, our methods can actively label uniformly distributed samples, demonstrating the ability to extract local information. However, other competitors tend to overly focus on difficult pixels, resulting in excessive and concentrated data collection in local boundaries and heterogeneous regions. An enlarged view (see Fig. 11) provides a clearer illustration of this advantage. In this view, the proposed GS-FusG variants only require the collection of a small number of scattered samples, while CASSL-MLR, PLCJ, and CNN-AL-MRF have a larger amount of labels.

B. Experiments With the WHU-Hi-HanChuan Dataset

The WHU-Hi-HanChuan dataset [62], [63] was gathered by the Headwall Nano-Hyperspec sensor, in Hanchuan, Hubei, China. It comprises an HSI with 274 bands and has a spatial resolution of approximately 0.109 m. The WHU-Hi-HanChuan image consists of 1217 × 303 pixels, 257 530 of which have reference labels divided into 16 classes, as shown in the table at the top of Fig. 12. In Fig. 12, there are many shadow-covered areas in the WHU-Hi-HanChuan scene, resulting in spectral distortion and variation, greatly complicating the classification task with a limited number of labeled samples. Therefore, the WHU-Hi-HanChuan dataset can be used to verify the effectiveness of the proposed method in scenarios with spectral distortion and variation.

In this experiment, we also evaluated the performance of our proposed method from the following two aspects.

1) Performance in Terms of AL Effectiveness: As shown in Fig. 13(a), with fewer than 132 labeled samples, GS-FusG[1] and GS-FusG[2] have a greater advantage over GS-FusG[3]. Notably, GS-FusG[3] has the fastest rate of improvement in classification performance. This reveals that, on the WHU-Hi-HanChuan data, compared with GS-FusG[1] and GS-FusG[2], GS-FusG[3] significantly raises the sampling effectiveness of GS and enhances the signal recovery ability using more labeled samples (when $N_q > 152$, the OA of GS-FusG[3] far exceeds that of GS-FusG[1] and GS-FusG[2]). Therefore, our proposed GS-FusG[3] is a very promising GS-based method, greatly enhancing the AL effectiveness, as shown in Fig. 13(a). In Fig. 13(b), it can be seen that, compared with other AL methods, our proposed GS-FusG also has the highest AL effectiveness and can achieve 90% classification accuracy when 132 samples are labeled.

2) Performance in Terms of Classification Accuracy: As shown in Table V, due to the influence of spectral variation, it is extremely difficult for existing AL algorithms to achieve satisfactory results when the number of active labels is less than 100. The best of them (CNN-AL-MRF) achieved OA, AA, and Kappa that are at least 7%, 9%, and 0.08 lower, respectively, than those obtained by our method. Among the variants of our proposed method, with 132 labeled samples, the OA of GS-FusG[1] exceeds 90%, while the AA of GS-FusG[2] is the highest (76.50%), reflecting that our newly proposed method can improve the classification performance of all categories in a balanced manner.

In Table VI, other comparison methods show poor classification stability across multiple categories (with standard deviations exceeding 10%), among which some AL approaches (CASSL-MLR, CNN-AL-MRF, and PLCJ) are severely affected by the *start-up* processes and their maximum instability even reached around 30%. In a more detailed experiment involving 62 labeled samples, we found that the standard deviations of OA for MI-MLR, BT-MLR, and CASSL-MLR reached 9%, 10%, and nearly 6%, respectively, demonstrating that the existing AL methods exhibit greater instability under extremely limited labels. Furthermore, in Table VI, NSCKL and GraphGST still have a significant gap compared with our methods. Using the same labeled samples, GS-FusG[3] outperformed GraphGST by a margin of 7% in OA. Although the proposed methods yield zero accuracies for Class 5 with 132 labeled samples, resulting in lower AAs than NSCKL, the impact is minimal due to the limited sample size of this category. Specifically, GS-FusG[3]'s zero accuracy originates

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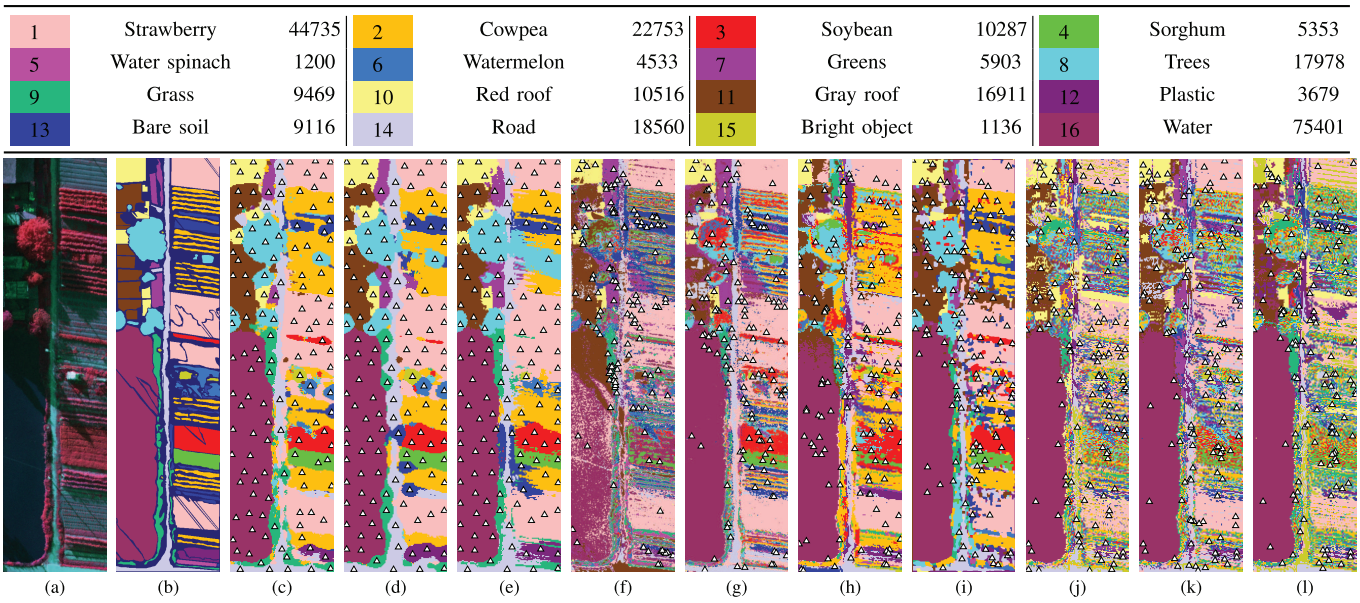


Fig. 12. Classification maps obtained by different AL algorithms (with 132 labeled samples illustrated in white triangles) applied to the WHU-Hi-HanChuan dataset. (a) False color composition. (b) Ground truth. (c) GS-FusG^1 (proposed). (d) GS-FusG^2 (proposed). (e) GS-FusG^3 (proposed). (f) MI-MLR. (g) CASSL-MLR. (h) PLCJ. (i) CNN-AL-MRF. (j) MC-MGR. (k) VOPT-MGR. (l) UNC-MGR.

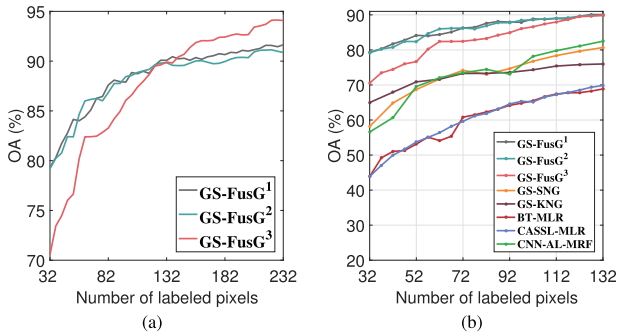


Fig. 13. OA curves obtained for the WHU-Hi-HanChuan dataset by (a) three variants of our proposed GS-based method and (b) all AL methods.

TABLE V

COMPARISON OF OA (%), AA (%), AND KAPPA OBTAINED BY DIFFERENT AL METHODS APPLIED TO THE WHU-HI-HANCHUAN DATASET WITH TOTAL 132 LABELED SAMPLES

	OA	AA	Kappa×100
MI-MLR	51.79 ± 3.51	43.16 ± 1.63	44.48 ± 3.22
BT-MLR	68.86 ± 2.65	53.59 ± 3.04	63.55 ± 2.57
CASSL-MLR	69.85 ± 3.53	54.52 ± 3.88	64.77 ± 3.84
PLCJ	69.62 ± 4.23	49.84 ± 5.30	64.62 ± 4.82
CNN-AL-MRF	82.49 ± 2.49	65.90 ± 4.55	79.40 ± 2.98
MC-MGR	60.27 ± 2.86	40.89 ± 3.40	53.93 ± 3.20
SOPT-MGR	64.45 ± 0.60	46.49 ± 1.32	58.67 ± 0.65
VOPT-MGR	66.53 ± 1.43	49.17 ± 2.50	61.12 ± 1.68
UNC-MGR	52.49 ± 3.24	38.73 ± 2.38	45.79 ± 3.16
GS-SNG	80.68	65.05	77.47
GS-KNG	75.99	53.36	71.65
GS-FusG^1	**90.08**	75.02	**88.35**
GS-FusG^2	89.84	**76.50**	88.08
GS-FusG^3	89.91	75.81	88.14

from its $N_{\text{cl}} = 154 > 132$. We further tested the case of 162 labeled samples and obtained the OA, AA, and Kappa of GS-FusG^3 as 92.06%, 83.14%, and 0.91, respectively, while the results of NSCKL were 82.17%, 82.12%, and 0.79, respectively. Thus, our method achieves approximately 10% higher OA while maintaining comparable AA, demonstrating its stronger focus on major land-cover categories and superior capability in handling class-imbalanced data.

In the obtained classification maps (see Fig. 12), we can see more clearly that the proposed GS-FusG^1, GS-FusG^2, and GS-FusG^3 were not disturbed by shadows in the WHU-Hi-HanChuan, and achieve the best results from the viewpoint of both local smooth structures and boundaries. However, the maps of MI-MLR, CASSL-MLR, PLCJ, MC-MGR, VOPT-MGR, and UNC-MGR exhibit obvious shapes and outlines of the shadow areas. In addition, for these areas, the proposed methods only require labeling a small number of pixels to reduce classification interference. However, other competitors often neglect to detect and query these areas or require extensive data collection. These results demonstrate that the proposed GS-FusG algorithm not only achieves high classification accuracies but also has strong resistance to spectral variability.

C. Experiments With the WHU-Hi-HongHu Dataset

The WHU-Hi-HongHu dataset [62], [63] was also collected by the Headwall Nano-Hyperspec sensor, in Honghu, Hubei, China. It comprises an HSI with 270 bands and has a spatial resolution of about 0.043 m. In order to further study the promotion of active classification in the case of very few labeled samples, we selected a 114 × 337 subset as shown in Fig. 14(a) for subsequent experiments. In this subregion, 34 169 pixels are labeled into 12 classes for reference, listed in the table at the top of Fig. 14. This dataset contains different classes of crops, as well as different cultivars of the same crop, which are extremely difficult to distinguish. Therefore, this dataset can be used to evaluate the classification ability of the proposed algorithm under limited labeled samples.

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TABLE VI

EXPERIMENTAL DETAILS (OA [%], AA [%], AND KAPPA) ON THE WHU-HI-HANCHUAN DATASET WITH TOTAL 132 LABELED SAMPLES

classNo.	Non-graph-based method		Graph-based method						
	CASSL-MLR	CNN-AL-MRF	PLCJ	VOPT-MGR	NSCKL	GraphGST	GS-FusG^1	GS-FusG^2	GS-FusG^3
1	93.14 ± 3.88	94.98 ± 2.17	85.53 ± 7.46	60.78 ± 5.79	78.15 ± 12.61	96.20	**98.67**	97.70	98.31
2	59.97 ± 11.98	78.43 ± 8.41	73.09 ± 13.19	26.08 ± 5.87	81.81 ± 6.95	57.60	**95.08**	84.89	71.84
3	52.69 ± 30.12	82.31 ± 10.31	62.46 ± 24.46	44.80 ± 11.21	83.67 ± 7.44	0.09	**85.89**	82.78	79.95
4	77.59 ± 23.67	76.04 ± 18.65	63.10 ± 17.34	35.13 ± 12.21	98.02 ± 1.47	92.82	**99.10**	96.32	93.93
5	28.91 ± 18.75	47.07 ± 33.56	14.65 ± 14.97	47.77 ± 17.63	**99.03** ± 2.18	0.00	0.00	0.00	0.00
6	15.30 ± 8.53	24.85 ± 13.61	12.04 ± 11.64	13.36 ± 8.46	**77.09** ± 8.11	16.07	29.10	41.28	31.15
7	83.99 ± 4.54	87.34 ± 3.67	71.91 ± 29.47	58.93 ± 16.83	87.95 ± 4.56	64.03	77.52	83.16	**95.31**
8	54.23 ± 12.66	68.90 ± 7.75	28.55 ± 27.33	21.31 ± 7.76	80.85 ± 2.30	85.89	88.05	92.39	**96.25**
9	47.03 ± 9.96	63.49 ± 12.19	36.24 ± 17.78	21.49 ± 11.37	68.55 ± 8.57	80.37	**84.05**	60.78	69.06
10	66.43 ± 32.30	75.86 ± 7.53	50.11 ± 11.23	72.55 ± 13.55	85.99 ± 5.39	95.96	95.00	**96.73**	93.75
11	52.82 ± 33.27	76.78 ± 15.94	63.13 ± 14.50	64.02 ± 21.15	69.57 ± 16.87	96.81	90.79	96.24	**98.26**
12	19.80 ± 18.63	39.43 ± 21.77	32.60 ± 22.43	31.46 ± 15.72	**88.93** ± 6.42	50.54	81.44	88.58	85.59
13	46.22 ± 8.69	37.88 ± 19.56	19.14 ± 13.80	18.41 ± 7.45	61.67 ± 5.17	**64.28**	26.72	54.80	64.18
14	67.65 ± 18.56	79.32 ± 5.74	64.05 ± 13.79	54.84 ± 12.08	60.54 ± 11.99	86.01	87.73	**88.41**	87.65
15	25.59 ± 16.41	24.40 ± 12.02	28.04 ± 26.52	54.98 ± 13.42	**77.39** ± 5.30	31.19	62.15	62.24	48.50
16	81.02 ± 17.16	97.39 ± 1.29	92.80 ± 3.59	89.53 ± 10.17	88.32 ± 5.52	98.66	98.96	97.75	**99.19**
OA	69.85 ± 3.53	82.49 ± 2.49	69.62 ± 4.23	66.53 ± 1.43	80.28 ± 3.30	82.93	**90.08**	89.84	89.91
AA	54.52 ± 3.88	65.90 ± 4.55	49.84 ± 5.30	49.17 ± 2.50	**80.47** ± 2.10	63.53	75.02	76.50	75.81
Kappa×100	64.77 ± 3.84	79.40 ± 2.98	64.62 ± 4.82	61.12 ± 1.68	77.32 ± 3.69	79.99	**88.35**	88.08	88.14

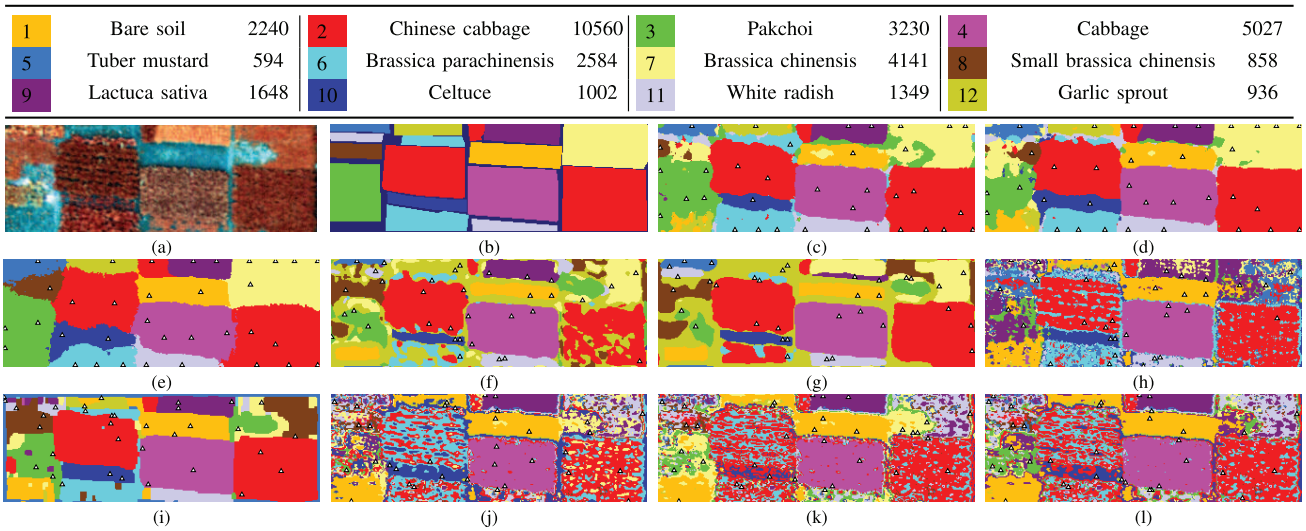


Fig. 14. Classification maps obtained by different AL algorithms (with 34 labeled samples illustrated in white triangles) when applied to the WHU-Hi-HongHu dataset. (a) False color composition. (b) Ground truth. (c) GS-FusG^1 (proposed). (d) GS-FusG^2 (proposed). (e) GS-FusG^3 (proposed). (f) MI-MLR. (g) CASSL-MLR. (h) PLCJ. (i) CNN-AL-MRF. (j) MC-MGR. (k) VOPT-MGR. (l) UNC-MGR.

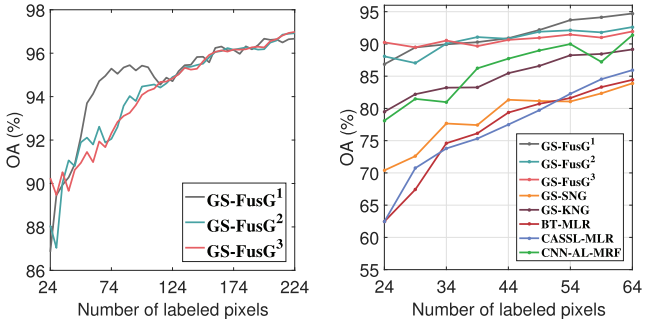


Fig. 15. OA curves obtained by different methods for the WHU-Hi-HongHu dataset.

Fig. 15 reveals that the AL performance of our proposed method is the best under a limited number of labels, and GS-FusG^3 reaches 90% accuracy with 24 labeled samples, while the deep learning-based method (CNN-AL-MRF) requires at least 54 labels. In addition, it can be seen that the traditional GS-SNG and GS-KNG methods exhibit poor classification accuracy in this case of label scarcity, which demonstrates that our proposed method can greatly increase the sampling effectiveness and the recovery precision of GS in HSI.

Table VII gives the detailed values of the three indicators. With a total of 34 labels, GS-FusG^3 exhibits a substantial AA enhancement, exceeding the single graph versions (GS-SNG and GS-KNG) by 10% and other AL techniques by 14%. In Table VIII, GS-FusG^3 outperforms NSCKL by 14% in OA and GraphGST by 10% in AA. Moreover, all other methods exhibit fluctuations in classification performance

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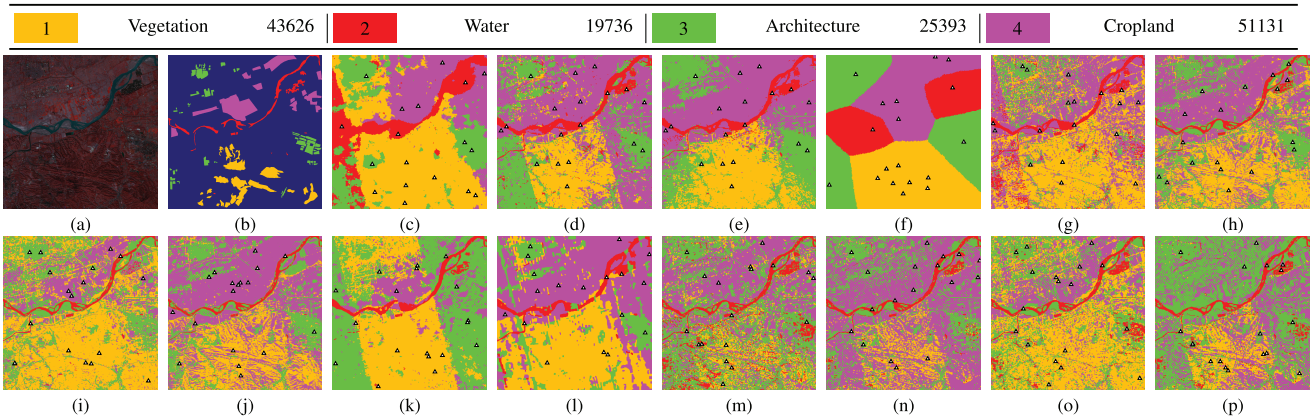


Fig. 16. Classification maps obtained by different AL algorithms (with 18 labeled samples illustrated in white triangles) when applied to the Wen-County dataset. (a) False color composition. (b) Ground truth. (c) GS-FusG^1 (proposed). (d) GS-FusG^2 (proposed). (e) GS-FusG^3 (proposed). (f) GS-SNG. (g) GS-KNG. (h) MI-MLR. (i) BT-MLR. (j) CASSL-MLR. (k) PLCJ. (l) CNN-AL-MRF. (m) MC-MGR. (n) VOPT-MGR. (o) SOPT-MGR. (p) UNC-MGR.

TABLE VII

COMPARISON OF OA (%), AA (%), AND KAPPA OBTAINED BY DIFFERENT AL METHODS FOR THE WHU-HI-HONGHU DATASET WITH TOTAL 34 LABELED SAMPLES

	OA	AA	Kappa×100
MI-MLR	69.91 ± 4.97	68.67 ± 3.57	65.39 ± 5.09
BT-MLR	74.60 ± 1.97	70.97 ± 3.97	70.26 ± 2.30
CASSL-MLR	73.79 ± 6.59	71.87 ± 4.73	69.64 ± 7.12
PLCJ	67.68 ± 3.44	55.40 ± 6.33	61.95 ± 3.95
CNN-AL-MRF	80.97 ± 1.32	73.36 ± 2.96	77.53 ± 1.49
MC-MGR	50.97 ± 2.65	43.12 ± 2.72	42.52 ± 2.19
SOPT-MGR	56.59 ± 2.47	47.24 ± 1.81	48.78 ± 2.68
VOPT-MGR	55.42 ± 2.32	45.76 ± 1.05	47.40 ± 2.06
UNC-MGR	50.99 ± 7.97	44.25 ± 1.95	43.20 ± 7.21
GS-SNG	77.66	64.49	72.99
GS-KNG	83.22	76.61	79.84
GS-FusG^1	89.94	83.02	88.02
GS-FusG^2	90.04	84.01	88.14
GS-FusG^3	**90.52**	**87.44**	**88.73**

due to the *start-up* process, especially on individual categories, and are noticeably inferior to our proposed methods. Accordingly, in Fig. 14, it can be seen that the classification map of GS-FusG^3 is also the most consistent with the ground truth, with well-delineated boundaries and smoothness. And the proposed GS-FusG methods can generate query points with a more reasonable distribution and greater representativeness, especially in some visually ambiguous areas.

D. Experiments on Wen-County Dataset

The Wen-County dataset [64] was acquired by the Gaofen-5 satellite in November 2019 over the Yellow River area near Wen County in Jiaozuo, Henan, China. The original HSI dataset contains 330 bands. We deleted 69 bands with a low signal-to-noise ratio. The final size of this HSI is $960 \times 960 \times 261$. We manually labeled a large number of samples from the dataset, including vegetation, water, architecture, and cropland, listed at the top of Fig. 16.

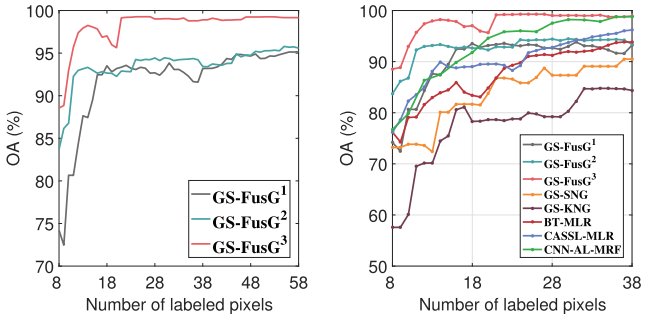


Fig. 17. OA curves obtained by different methods on the Wen-County dataset.

In Fig. 17, it can be seen that the proposed GS-FusG method greatly improves the AL effectiveness of traditional GS-based methods. What is more noteworthy is that GS-FusG^3 significantly increases the AL upper limit of GS-FusG^1 and GS-FusG^2 (from 95% to 99%). Specifically, GS-FusG^3 achieves OA stability at 99% with 21 labels, while CNN-AL-MRF requires more than 38 labels. Moreover, in Table IX, with 18 labeled samples, GS-FusG^3 is the only approach maintaining >90% accuracy across all classes while achieving 97.02% OA—surpassing CNN-AL-MRF by 5.24% and others by over 8%. Among them, the accuracies of benchmark AL methods also carry large standard deviations, reflecting unstable performance. By contrast, our proposed GS-FusG can obtain deterministic high-precision classification results.

In Fig. 16, our method can obtain remarkable segmentation results for categories such as architecture, water, and vegetation, among which the classification map of GS-FusG^3 not only contains abundant detailed boundary information but also exhibits high local smoothness. However, other competitors produce either very fragmented results (GS-KNG, MI-MLR, BT-MLR, CASSL-MLR, MC-MGR, VOPT-MGR, SOPT-MGR, and UNC-MGR) or over-smoothing effects between different land covers (PLCJ, CNN-AL-MRF, and MC-MGR).

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TABLE VIII

EXPERIMENTAL DETAILS (OA [%], AA [%], AND KAPPA ON THE WHU-HI-HONGHU DATASET WITH TOTAL 34 LABELED SAMPLES

classNo.	Non-graph-based method		Graph-based method						
	CASSL-MLR	CNN-AL-MRF	PLCJ	VOPT-MGR	NSCKL	GraphGST	GS-FusG[1]	GS-FusG[2]	GS-FusG[3]
1	75.10 ± 18.48	72.80 ± 35.79	85.66 ± 19.51	63.78 ± 36.54	76.73 ± 25.37	84.10	80.62	**95.40**	93.88
2	85.06 ± 21.00	96.42 ± 0.53	86.76 ± 6.48	74.91 ± 9.42	64.06 ± 18.78	92.26	98.13	**98.63**	93.22
3	40.30 ± 12.87	69.06 ± 11.22	31.39 ± 10.44	30.25 ± 16.07	54.48 ± 10.54	42.94	90.79	68.17	**96.34**
4	94.81 ± 4.51	97.74 ± 2.82	98.19 ± 0.84	86.71 ± 2.25	97.74 ± 1.11	98.75	99.00	**99.54**	98.21
5	63.93 ± 19.94	58.94 ± 30.20	17.94 ± 13.07	24.39 ± 15.42	95.13 ± 1.47	38.11	92.41	88.53	**98.82**
6	66.46 ± 27.95	82.49 ± 4.00	58.81 ± 7.67	16.76 ± 9.94	83.71 ± 6.85	84.08	**89.07**	72.84	61.14
7	46.69 ± 16.39	46.01 ± 18.02	29.79 ± 13.89	20.22 ± 7.58	74.82 ± 8.47	86.92	81.77	95.62	**100.00**
8	68.83 ± 24.80	68.13 ± 36.94	26.05 ± 27.97	35.19 ± 8.27	81.36 ± 16.69	65.11	27.42	43.76	**82.15**
9	68.02 ± 26.56	**94.91** ± 4.96	83.45 ± 7.49	77.62 ± 6.04	94.82 ± 1.21	92.71	88.34	91.25	80.32
10	87.78 ± 5.30	93.48 ± 4.11	53.78 ± 27.58	44.99 ± 4.54	96.26 ± 6.76	79.52	85.91	**99.40**	93.50
11	75.05 ± 10.17	72.21 ± 10.77	54.37 ± 11.19	27.99 ± 11.22	70.66 ± 10.52	**86.71**	82.33	75.65	58.20
12	90.43 ± 10.69	28.10 ± 19.32	38.61 ± 25.98	46.35 ± 26.14	89.73 ± 13.76	77.86	80.43	79.36	**93.48**
OA	73.79 ± 6.59	80.97 ± 1.32	67.68 ± 3.44	55.42 ± 2.32	76.09 ± 6.11	84.16	89.94	90.04	**90.52**
AA	71.87 ± 4.73	73.36 ± 2.96	55.40 ± 6.33	45.76 ± 1.05	81.63 ± 2.67	77.42	83.02	84.01	**87.44**
Kappa×100	69.64 ± 7.12	77.53 ± 1.49	61.95 ± 3.95	47.40 ± 2.06	72.53 ± 6.73	81.31	88.02	88.14	**88.73**

TABLE IX

EXPERIMENTAL DETAILS (OA [%], AA [%], AND KAPPA) ON THE WEN-COUNTY DATASET WITH TOTAL 18 LABELED SAMPLES

classNo.	Non-graph-based method		Graph-based method						
	CASSL-MLR	CNN-AL-MRF	PLCJ	VOPT-MGR	NSCKL	GraphGST	GS-FusG[1]	GS-FusG[2]	GS-FusG[3]
1	92.97 ± 6.63	98.42 ± 1.63	89.97 ± 6.85	69.73 ± 28.02	93.23 ± 12.41	95.35	**99.03**	84.12	92.73
2	84.98 ± 10.79	91.69 ± 10.36	88.79 ± 12.03	95.10 ± 6.86	86.44 ± 8.72	99.13	97.05	99.74	**99.77**
3	84.38 ± 14.34	84.19 ± 11.94	84.37 ± 11.84	55.52 ± 30.49	91.43 ± 6.85	67.78	**96.53**	94.95	96.24
4	89.48 ± 10.27	89.93 ± 9.63	98.59 ± 1.20	71.40 ± 22.68	84.30 ± 16.62	88.04	85.94	96.06	**100.00**
OA	89.01 ± 3.90	91.78 ± 3.24	86.85 ± 8.65	71.34 ± 6.94	88.68 ± 7.29	88.21	93.51	92.65	**97.02**
AA	87.95 ± 3.14	91.06 ± 3.15	86.24 ± 8.31	72.94 ± 3.96	88.85 ± 6.39	87.57	94.64	93.71	**97.19**
Kappa×100	84.64 ± 5.34	88.52 ± 4.48	81.84 ± 11.80	60.39 ± 9.00	84.13 ± 10.19	83.33	91.00	89.74	**95.85**

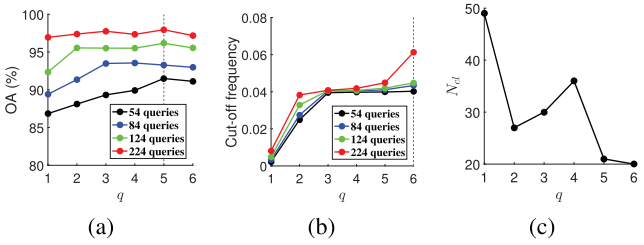


Fig. 18. Effect of parameter q on GS-FusG in terms of (a) classification accuracy, (b) cutoff frequency, and (c) sampling effectiveness.

V. DISCUSSION

Here, we further analyze the parameter influence of the proposed GS-FusG algorithm, its comprehensive performance in terms of efficiency, accuracy, and storage, and discuss the impact of graph structure on the optimality of GS.

A. Parameter Analysis

In (7), increasing q gives a more accurate estimation of the cutoff frequency $\Omega_q(\mathcal{S})$ and theoretically improves the sampling effectiveness of GS. In the previous HSI classification experiments, considering the computational efficiency, we only used $q = 2$.

In Fig. 18, we analyzed the effect of parameter q on classification accuracy, cutoff frequency, and the size of minimum complete set (N_{cl}), respectively, using the Indian Pines data. The algorithm used in Fig. 18 is GS-FusG[1]. When $q = 5$, the OA exceeds 90% with 54 labeled samples, which is higher than that of $q = 2$ adopted in Table III and Fig. 10. As can be seen from Fig. 18(a), with the increase of q, the overall classification performance under different label quantities also becomes better. Fig. 18(b) shows that, with the increase of q, the cutoff frequency $\Omega_q(\mathcal{S})$ also increases. Moreover, when $q = 6$, $\Omega_q(\mathcal{S})$ increases significantly with 224 labeled samples. According to (8), the more $\Omega_q(\mathcal{S})$ increases, the higher the sampling effectiveness of GS, and the more perfect signal reconstruction can be achieved. In Fig. 18(c), when q increases to 5 or 6, N_{cl} decreases significantly. This also indicates that the larger the q value makes our proposed GS-FusG algorithm have higher AL effectiveness in HSI classification (requiring fewer labeled samples to obtain the minimum complete set). Therefore, using a larger q, the advantages of our proposed GS-FusG algorithm become more remarkable compared with previous experimental results.

B. Computational Efficiency–Accuracy–Memory Co-Analysis

We comprehensively compared the computational efficiency and classification accuracy of several AL methods. In Fig. 19, the proposed GS-FusG has moderate query time but significantly superior OAs on four tested HSIs. In addition, as illustrated in Fig. 5, most of the existing AL algorithms require interrupting the sampling process, followed by field exploration, annotation, model updating, and then restarting the next

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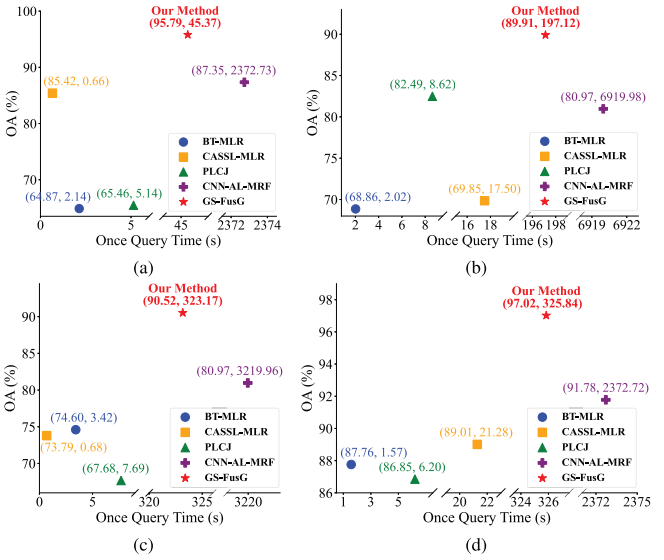


Fig. 19. "Time-Accuracy" comprehensive comparisons of AL methods. (a) Indian Pines. (b) WHU-Hi-HanChuan. (c) WHU-Hi-HongHu. (d) Wen-County.

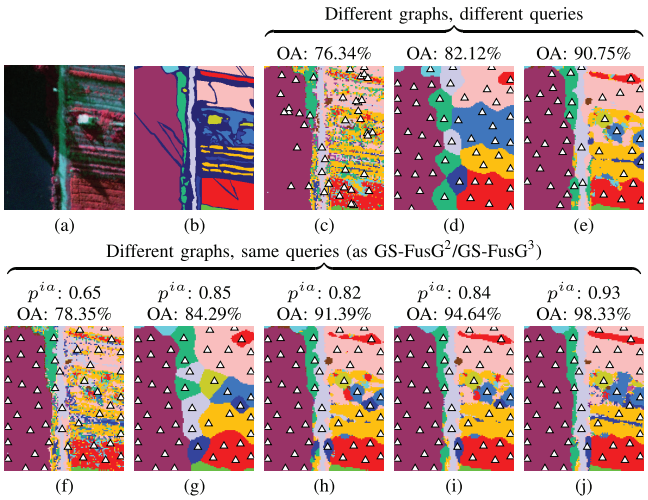


Fig. 20. Influence of graph structure on global optimality (with 162 labeled samples on WHU-Hi-HanChuan). (a) False color composition. (b) Ground truth. (c) GS-KNG. (d) GS-SNG. (e) GS-FusG^1. (f) KNG. (g) SNG. (h) FusG. (i) FusG-R-I. (j) FusG-R-II.

TABLE X
MEMORY USAGE OF FUSG AND ITS KNG COMPONENT

	Indian Pines	WHU-Hi-HanChuan	WHU-Hi-HongHu	Wen-County
KNG	4.81MB	11.14MB	9.69MB	8.15MB
FusG	16.78MB	36.16MB	35.65MB	31.47MB

sampling/querying. While the sampling process of GS-FusG and PLCJ is a one-time operation, it ensures the continuity of subsequent annotation and classification, thus demonstrating higher execution efficiency in real AL flow. Nevertheless, the samples queried by PLCJ have poor representativeness, resulting in low OAs on most HSIs.

In addition, FusG employs sparse graph storage and superpixel-based graph vertices, which significantly reduces memory usage. In Table X, for large-sized HSI data, the memory usage of FusG is less than 40 MB, being about 4 times that of its KNG component.

Therefore, the proposed GS-FusG demonstrates excellent comprehensive performance: moderate query time, model-free sampling process, high accuracy, and low memory usage.

C. Optimality Analysis of Graph Structure

The traditional GS can only sample the "globally optimal" points within the relationships modeled by the current graph, thus ignoring the influence of the graph structure on the optimality. In this article, we demonstrate the optimal graph theory for sampling and reconstruction, and propose the GS-FusG methods based on multiview graphs, specifically tailored for multidimensional graph signals in the real world. However, in the context of HSI with spectral variability and noise, it is impossible to obtain an idealized graph structure.

We discuss this issue by referring to the shadow area in Fig. 20. The first line explains that different graph structures will result in divergent query distributions, in which the proposed GS-FusG can more keenly explore the complex and informative regions and obtain their more precise predictions than the commonly used KNG and SNG. Specifically, GS-FusG is less affected by shadows in the *Soybean* (red) and *Bright object* (claybank) areas. However, GS-KNG fails to resist the influence of shadows on the classification, while GS-SNG fails to selectively sample specific shadow-affected areas and generates distorted boundaries.

The second line of Fig. 20 compares the performance of various graph structures with the same query points. We define p^{ia} as the proportion of the intraclass graph weights

$$p^{ia} = \frac{\sum_{y_i=y_j} W_{ij}}{\sum_{y_i=y_j} W_{ij} + \sum_{y_i\neq y_j} W_{ij}}. \tag{39}$$

FusG-R-I/II denotes the enhanced graph obtained by removing the interclass connections (i.e., reducing $\sum_{y_i\neq y_j} W_{ij}$) in FusG referring to the ground truth. Using 162 labeled samples queried by GS-FusG^2, FusG attains over 90% OA with a robust classification map, especially in shadow-affected regions, substantially exceeding the performance of KNG and SNG. By further reducing interclass connections of FusG, the obtained FusG-R-I/II gradually approaches the idealized graph, with a larger p^{ia} value, and the generated classification map is increasingly closer to the ground truth. However, under limited prior knowledge, FusG-R-I/II is agnostic.

The above analyses indicate that the GS technology based on the idealized graph can query the globally optimal samples and reconstruct the optimal graph signal in the real HSI scene, rather than merely calculating the globally optimal vertices in a specific graph. Moreover, our proposed GS-FusG methods show promising capability to approximate the sampling effectiveness and the recovery precision of the idealized graph, with extremely limited labeled data.

VI. CONCLUSION

In this article, we have designed a new AL algorithm based on sampling and reconstruction theory in graph signal

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processing for HSI classification. Our new algorithm fuses multiview connection information. Compared with existing AL methods, our newly developed GS-FusG exhibits important advantages, such as higher AL effectiveness and better classification performance with limited labels. The GS-FusG^3 variant is more suitable for multidimensional graph signals and can increase the upper limit of GS-based methods. For HSIs with severe spectral variations, our method can greatly improve the smoothness of the graph signal and obtain classification results with enhanced boundaries and local consistency. As with any new work, our method still has some aspects to be improved, including high computational complexity on large-sized HSIs and fixed graph structure. In future work, we will further address the problem of computational efficiency and optimize the graph structure in GS by dynamic graph learning technology concepts.

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